



scverse conference 2025

Stanford, CA

November 17 – 19

Abstract Book

Conference Schedule

Day 01 — Monday, November 17

Location: LKSC

08:00	<i>Break</i>	Breakfast
09:00	Talk	Welcome Session — <i>Emma Dann (scverse)</i>
09:15	Talk	State of scverse — <i>Mikaela Koutrouli (scverse)</i>
09:45	Keynote	John Marioni — <i>Chair: Emma Dann (scverse)</i>
10:30	<i>Break</i>	Coffee
11:00	Talk	Clonally resolved Perturb-seq uncovers lineage-level responses to genetic perturbation — <i>Dillon Lue (Princeton)</i>
11:15	Talk	PerTurbo: A scalable Bayesian framework for whole-transcriptome differential expression analysis and optimal design of single-cell CRISPR screens — <i>Logan Blaine (Harvard)</i>
11:30	Talk	ChronoSeq: minute scale automated scRNA-seq for real time profiling of perturbations — <i>Kanishk Asthana (UCSD)</i>
11:45	Talk	Sponsor Talk: Towards a virtual cell signaling model rooted in human development — <i>Samantha Dale Strasser (Somite Therapeutics)</i>
12:00	<i>Lunch</i>	Lunch
13:30	Keynote	Panos Roussos — <i>Chair: Mikaela Koutrouli</i>
14:15	Talk	Universal MPRA: Zero-shot regulatory element activity prediction across tissue and cell types — <i>Laura Gunsalus (Genentech)</i>
14:30	Talk	LIVI: An interpretable deep learning framework for mapping genetic effects at cellular resolution using population-scale single-cell RNA sequencing data — <i>Danai Vagiaki (EMBL)</i>
14:45	Talk	Attention-based multiple instance learning predicts CAR T cell therapy outcomes from infusion product single-cell RNA-seq data and identifies engineering targets in large B cell lymphoma — <i>Kameron Rodrigues (Stanford)</i>
15:00	<i>Break</i>	Coffee
15:30	Panel	Agentic Workflows for Bioinformatics — <i>Jonah Cool (Anthropic), Vega Shah (NVIDIA), Loic Royer (CZ Biohub), Kexin Huang (Stanford). Chair: Giovanni Palla (scverse)</i>
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Location: LKSC

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10:00	Talk	SPLISOSM: Unlocking hidden isoform landscapes in spatial transcriptomics data — <i>Jiayu Su (Columbia)</i>
10:15	Talk	Robust integration of sparse single-cell alternative splicing and gene expression data with SpliceVI — <i>Smriti Vaidyanathan (Columbia)</i>
10:30	Talk	Sponsor Talk: Powering the Virtual Cell: Frontier Datasets for AI-Driven Biology — <i>Johnny Yu (Tahoe Therapeutics)</i>
10:45	Poster	Poster Session 2
12:00	<i>Lunch</i>	Lunch
13:30	Keynote	Elham Azizi — <i>Chair: Pau Badia (scverse)</i>
14:15	Talk	X-Atlas/Orion: Genome-wide Perturb-seq datasets via a scalable fix-cryopreserve platform for training dose-dependent biological foundation models — <i>Ann Huang (Xaira)</i>
14:30	Talk	Decoding sequence determinants of gene expression across cellular and disease states — <i>Avantika Lal (Genentech)</i>
14:45	Talk	MACS3: A versatile peak caller for single-cell genomics — <i>Tao Liu (Roswell Park)</i>
15:00	Talk	Sponsor Talk: Solving complex challenges in human health at the intersection of AI and biology — <i>Angela Oliveira Pisco (CZI)</i>
15:15	<i>Break</i>	Coffee
15:45	Talk	grassp: An integrated framework for graph-based analysis of spatial/subcellular proteomics — <i>Max Frank (CZ Biohub)</i>
16:00	Talk	InterSCellar: surface-based cell interaction volume analysis in 3D spatial omics — <i>Eunice Lee (Harvard)</i>
16:15	Talk	EasyVitesce: Adding interactivity to scverse single-cell and spatial biology plots — <i>Eric Mörth (Harvard)</i>
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Clonally Resolved Perturb-seq Uncovers Lineage-level Responses to Genetic Perturbation

Dillon Lue

Monday, 11:00

Single-cell RNA-seq experiments with CRISPR knockouts (Perturb-seq) enable systematic dissection of gene function by measuring perturbation effects at single-cell resolution. Single-cell RNA-seq experiments paired with clonal assays such as TCR/BCR sequencing or lineage-tracing approaches reveal clonal expansion and diversity. We present a clonal barcoding extension to Perturb-seq that tags each transduced cell with a heritable barcode linked to its guide RNA. As a result, all descendants of a progenitor cell inherit both the same perturbation and the same barcode, allowing us to track clonal expansion dynamics and cell state composition for an entire clone for a given genetic perturbation.

We apply this framework to tumor-infiltrating CD8 T cells in vivo in mice, a system directly relevant to immunotherapy where maintaining stem-like populations within expanding clones is critical for long-term tumor control. With clonally barcoded Perturb-seq, we (i) classify clonal heterogeneity across control and perturbed cells, (ii) detect perturbation effects more robustly than cell-only analyses, (iii) identify novel cell states formed selectively by a few clones and (iv) reconstruct the order of differentiation from stem-like progenitors to more differentiated effector states. Beyond this case study, the approach generalizes to other systems, offering a versatile framework for studying how genetic perturbations shape clonal dynamics, lineage decisions, and fate hierarchies.

PerTurbo: A scalable Bayesian framework for whole-transcriptome differential expression analysis and optimal design of single-cell CRISPR screens

Logan Blaine
Monday, 11:15

Pooled single-cell CRISPR screens have revolutionized functional genomics by enabling large-scale perturbation studies of the genome. Yet current computational analysis approaches suffer from critical limitations. Existing fast statistical approaches produce many false positives, but more robust alternatives are markedly slower, making accurate transcriptome-wide testing for differentially expressed genes challenging.

To address these challenges, we developed PerTurbo: a scalable, fully Bayesian analysis framework for large-scale Perturb-seq experiments. PerTurbo uses GPU-accelerated stochastic variational inference, implemented with scvi-tools and Pyro, to efficiently analyze perturbation effects in datasets comprising millions of cells and thousands of perturbations. Crucially, unlike previous statistical tools, it also models CRISPR guide efficiency as a source of technical variability, allowing more accurate effect size estimation. We benchmarked PerTurbo using simulated datasets, where it performed competitively with existing methods in accurately identifying differentially expressed genes. To assess real-world performance, we reanalyzed an enhancer-mapping screen and found that PerTurbo identified similar enhancer-gene pairs compared with existing robust methods, while running $>10\times$ faster. This enabled us to exhaustively test for trans effects on distal genes, revealing enhancers of GATA1 and LMO2 whose repression affected hundreds of genes. To underscore PerTurbo's scalability, we next reanalyzed a genome-wide Perturb-seq screen with ~ 2 million cells in under two hours on a single machine, replicated key findings from the original analysis, and detected effects for "weak" perturbations overlooked in the previous analysis. Finally, we developed a simulation-based power analysis tool to help determine optimal sequencing depth and cell requirements for a given budget and experimental design.

ChronoSeq: minute scale automated scRNA-seq for real time profiling of perturbations

Kanishk Asthana

Monday, 11:30

Current single-cell RNA sequencing technologies achieve temporal resolution of ~ 30 minutes at best, missing rapid transcriptional changes that occur within minutes of cellular perturbations. We developed ChronoSeq, a novel droplet-based technology enabling unprecedented 7-minute temporal resolution for single-cell transcriptomics through innovative time-tagged DNA barcoded beads and automated microfluidic workflows.

ChronoSeq incorporates time-specific sequence tags into modified Drop-seq beads, coupled with a blockage-resistant microfluidic device supporting 12 unique time-tags across extended 2+ hour experiments. We applied ChronoSeq to profile inflammatory responses in K562 cells stimulated with TNF-, capturing the rapid NF- κ B signaling cascade previously undetectable at conventional temporal scales. Our data revealed distinct activation kinetics among inflammatory genes. Application of Optimal Transport analysis reveal a gradient of activation depending on differentiation towards Erythroid phenotype.

Additionally, we performed mixed-cell perturbation experiments using K562 and differentiated THP-1 cells with LPS stimulation to examine cell-type-specific inflammatory responses and Cell-To-Cell communication dynamics. Moreover, to dissect transcriptional versus translational dynamics, we conducted puromycin translation-blocking experiments following TNF- stimulation in K562 cells, revealing pure transcriptional kinetics without translational feedback.

ChronoSeq maintains high data quality with $<5\%$ doublet rates while dramatically improving temporal resolution. This technology enables precise mapping of perturbation-induced transcriptional cascades, inference of regulatory causality, and characterization of cell-type-specific response dynamics previously inaccessible to single-cell genomics, opening new avenues for understanding cellular perturbation responses.

Universal MPRA: Zero-shot regulatory element activity prediction across tissue and cell types

Laura Gunsalus
Monday, 14:15

Understanding regulatory element activity across human cell types remains a central challenge in functional genomics, as experimental assays like MPRA are limited to few cell lines and computational models fail to generalize beyond their training contexts. We present Universal MPRA, a computational framework that leverages long-context DNA-to-expression models trained on millions of genomic loci to enable zero-shot prediction of regulatory element activity in any cell type, including primary tissues and rare cell types inaccessible to experimental assays. The approach generates in silico reporter constructs and predicts cell-type-specific expression using models like Borzoi and Decima. Through systematic benchmarking, we demonstrate high concordance with experimental MPRA measurements in HepG2 and K562 cells ($r = 0.78-0.79$), successfully predict tissue-specific activity for promoters like brain-specific hSyn, and identify unexpected tissue biases in ubiquitous promoters like CAG and CMV. This framework enables flexible, model-agnostic analysis of regulatory constructs across primary tissues and rare cell types where experimental measurement remains impossible, providing a general-purpose surrogate for cell-type-specific regulatory assays.

LIVI: An interpretable deep learning framework for mapping genetic effects at cellular resolution using population-scale single-cell RNA sequencing data

Danai Vagiaki
Monday, 14:30

Expression quantitative trait loci (eQTL) studies can elucidate the functional consequences of disease-associated genetic polymorphisms. While scRNA-seq technologies applied to population-scale cohorts enable eQTL discovery at single-cell resolution, effective strategies for trans-eQTL mapping at the single-cell level are lacking. Existing methods rely on predefined cell types, often aggregating single-cell counts into pseudocell clusters. They typically utilize single-gene models, imposing multiple testing challenges and failing to capture the complexity of gene regulatory networks (GRNs), wherein a variant may impact multiple downstream genes in trans. To address these challenges, we developed LIVI, an interpretable deep learning framework for efficient mapping of trans effects at cellular resolution. LIVI utilizes amortized variational inference to integrate single-cell expression counts from multiple individuals, experiments, and cell types, without requiring predefined cell types. Gene expression variation is decomposed into cell state variation and cell state-specific donor effects, while accounting for batch effects and global interindividual variation. LIVI identifies gene sets within specific subsets of cells, affected by genetics or other donor-level variables in a data-driven manner. Application of LIVI to the largest public scRNA-seq dataset of genetically diverse human donors replicated many known trans-acting variants from bulk meta-analyses, outperforming alternative latent variable methods. Moreover, LIVI identified novel trans-eQTLs in continuous cell states and finer-grained populations than pre-annotated cell types, as well as trans effects at the GRN level, missed by conventional single-gene models. To demonstrate utility for studying disease mechanisms, we mapped polygenic risk scores for autoimmune diseases and recovered many established disease-gene and disease-cell type associations.

Attention-based multiple instance learning predicts CAR T cell therapy outcomes from infusion product single-cell RNA-seq data and identifies engineering targets in large B cell lymphoma

Kameron Rodrigues

Monday, 14:45

Predictive modeling using single-cell RNA sequencing remains limited in clinical settings where outcomes are defined at the sample level. This problem is especially acute in synthetic contexts like chimeric antigen receptor (CAR) T cell therapy, where infused cellular products vary dramatically across patients, lie outside the training distributions of existing single-cell foundation models, and often possess batch effects. We introduce tcellMIL, a biologically informed multiple instance learning (MIL) framework that models each patient sample as a collection of unlabeled cells to identify influential cell populations and states to predict therapeutic response. tcellMIL incorporates prior biological knowledge by leveraging SCENIC, a gene regulatory network inference method using known transcription factor binding motifs to compute 164 regulon activity scores — biologically grounded features that reduce dimensionality and batch effects ($P = 2.87 \times 10^{-5}$). These features are denoised via a self-supervised autoencoder and combined with encodings such as batch or clinical covariates to improve cross-cohort generalization and multi-modal support. An attention-based MIL mechanism identifies the most outcome-relevant subpopulations, providing interpretability at cell and regulon levels. Applied to 64 CD19-directed CAR T cell infusion products across 3 patient cohorts, using leave-one-patient-out cross-validation, tcellMIL significantly outperformed pseudobulk and standard MIL baselines at predicting patient response (acc = 0.72 vs. 0.53-0.69; F1 score: 0.74 vs. 0.58-0.73), and identified regulatory programs — such as TBX21 and FOSB — and populations that drive therapeutic outcomes. Our results highlight a generalizable path for outcome prediction from scRNA-seq data where labels exist only at the sample level and cellular distributions deviate from standard atlases. This approach offers a pathway toward precision CAR T cell therapy design through predictive modeling and evidence-based cellular engineering strategies.

SPLISOSM: Unlocking Hidden Isoform Landscapes in Spatial Transcriptomics Data

Jiayu Su

Tuesday, 10:00

Modern spatial transcriptomics (ST) platforms inherently capture some isoform information — whether through 3' fragments in short-read approaches, full-length transcripts in long-read sequencing, or exon probes in in situ platforms. Yet most analysis pipelines discard this rich transcript diversity by aggregating to gene-level expression. We present SPLISOSM (Spatial ISOform Statistical Modeling), a computational framework that recovers this hidden layer of spatial complexity and its regulation through theoretically grounded statistical tests.

SPLISOSM reformulates spatial variability detection and differential isoform usage analysis as multivariate independence testing. Leveraging kernels optimized for spatial and compositional data, it demonstrates provably higher statistical power and robustness in simulation and real data. Implemented in Python, SPLISOSM seamlessly extends existing gene-level workflows to isoform-resolution. It is compatible with major ST platforms (e.g. 10X Visium, Visium + ONT, 10X Xenium Prime 5K), with memory-efficient low-rank approximations enabling spatial variability testing of hundreds of thousands of spots in minutes on a standard laptop.

Reanalysis of public mouse brain ST datasets identified over 1,000 spatially variable transcript diversity events, primarily in synaptic signaling pathways linked to neuropsychiatric disorders. We further uncovered evolutionarily conserved patterns between mouse and human prefrontal cortex, with region-specific RNA-binding proteins orchestrating local isoform regulation. In human glioblastoma, SPLISOSM exposed microenvironment-driven transcript switching in immune and adhesion pathways with direct therapeutic implications.

There can be no true "transcriptomics" without transcript-level resolution. SPLISOSM empowers researchers to revisit existing datasets and extract new biological insights from the isoform information that was always there. Link to preprint: <https://doi.org/10.1101/2025.05.02.651907>

Robust Integration of Sparse Single-Cell Alternative Splicing and Gene Expression Data with SpliceVI

Smriti Vaidyanathan
Tuesday, 10:15

Alternative splicing (AS) is a sparse yet informative signal in single-cell RNA-seq capturing cell-state heterogeneity beyond total gene expression (GE). While multimodal variational auto-encoders (VAEs) (e.g., MultiVI) integrate GE with other assays, joint modeling of AS and GE signals in single cells remains unexplored. We present SpliceVI, a multimodal VAE built on the scvi-tools framework and inspired by MultiVI's modality-specific encoders/decoders. We also curate a mouse "foundation" Smart-seq2 dataset for development and benchmarking. The AS arm uses a partial encoder (inspired by the EDDI framework) that conditions on only observed junctions, and the GE arm uses the standard scVI encoder. The splicing input comprises per-junction usage ratios (PSI) for observed junctions, which are paired with learnable per-junction embeddings. These are then passed through a shared layer and pooled into a cell-level splicing representation that feeds an MLP encoder. Modality latent spaces are concatenated to form a joint representation, and both modalities use linear decoders for interpretability. Training uses amortized variational inference with KL warm-up. The splicing decoder maximizes a Dirichlet-Multinomial over each observed alternative transcript structure event (ATSE)'s junction counts, and the GE decoder maximizes a zero-inflated negative binomial (ZINB) over gene counts. In the trained model, each modality-specific component of the joint embedding is uniquely informative. The joint latent yields the strongest performance on a variety of tasks, exceeding its single-modality halves and single-modality VAE baselines, and the splicing decoder reliably imputes held-out splicing signal, suggesting promise for unified GE and AS modeling.

X-Atlas/Orion: Genome-wide Perturb-seq Datasets via a Scalable Fix-Cryopreserve Platform for Training Dose-Dependent Biological Foundation Models

Ann Huang

Tuesday, 14:15

The rapid expansion of massively parallel sequencing technologies has enabled the development of foundation models to uncover novel biological findings. While these have the potential to significantly accelerate scientific discoveries by creating AI-driven virtual cell models, their progress has been greatly limited by the lack of large-scale high-quality perturbation data, which remains constrained due to scalability bottlenecks and assay variability. Here, we introduce “Fix-Cryopreserve-ScRNAseq” (FiCS) Perturb-seq, an industrialized platform for scalable Perturb-seq data generation. We demonstrate that FiCS Perturb-seq exhibits high sensitivity and low batch effects, effectively capturing perturbation-induced transcriptomic changes and recapitulating known biological pathways and protein complexes. In addition, we release X-Atlas: Orion edition (X-Atlas/Orion), the largest publicly available Perturb-seq atlas. This atlas, generated from two genome-wide FiCS Perturb-seq experiments targeting all human protein-coding genes, comprises eight million cells deeply sequenced to over 16,000 unique molecular identifiers (UMIs) per cell. Furthermore, we show that single guide RNA (sgRNA) abundance can serve as a proxy for gene knockdown (KD) efficacy. Leveraging the deep sequencing and substantial cell numbers per perturbation, we also show that stratification by sgRNA expression can reveal dose-dependent genetic effects. Taken together, we demonstrate that FiCS Perturb-seq is an efficient and scalable platform for high-throughput Perturb-seq screens. Through the release of X-Atlas/Orion, we highlight the potential of FiCS Perturb-seq to address current scalability and variability challenges in data generation, advance foundation model development that incorporates gene-dosage effects, and accelerate biological discoveries.

Decoding sequence determinants of gene expression across cellular and disease states

Avantika Lal

Tuesday, 14:30

Single-cell expression measurements have profiled gene expression patterns across diverse cell types and states. However, current approaches to reveal gene regulatory mechanisms from such data are limited, often requiring paired scATAC-seq data or prior knowledge of transcription factors. In contrast, recent sequence-to-function deep learning models have demonstrated that the genomic sequence itself offers a rich and largely untapped data modality for uncovering regulatory mechanisms. However, current state-of-the-art models have been trained largely on bulk expression profiles from healthy tissues or cell lines. Thus, they cannot predict expression in specific cell types. We present Decima, a sequence-to-function model trained on atlas-scale single-cell expression data. Decima is based on the Borzoi model (Linder et al. 2025), and is trained on pseudobulked scRNA-seq data from over 22 million cells, enabling predictions in specific cell types and conditions. Given the 500 kb sequence surrounding an unseen gene, Decima can predict the expression level of that gene in hundreds of cell types, across diverse cellular states, tissues, and disease conditions. Decima uncovers regulatory elements as well as motifs for transcription factors that define cell type identities. It successfully predicts subtle changes in gene expression between the same cell type across different tissues, states, or diseases, revealing the underlying regulatory mechanisms. In addition, it successfully predicts the impact of non-coding genetic variants at pseudobulk resolution. We show that Decima achieves state-of-the-art performance in identifying variants that alter gene expression, predicting their effects, and suggesting mechanistic hypotheses. This is vital for interpreting GWAS variants, over 90% of which are non-coding. Finally, Decima can design regulatory elements that drive cell type specific expression, which is crucial for developing targeted gene therapies.

MACS3: a versatile peak caller for single-cell genomics

Tao Liu

Tuesday, 14:45

MACS is one of the most widely adopted and highly cited peak callers for regulatory element discovery, supporting large-scale projects such as ENCODE and the Human Cell Atlas. In its latest release, MACS3 brings major performance improvements in memory usage and speed and has been integrated into MAESTRO, enabling seamless use in multi-modal single-cell transcriptome-regulome integration workflows. MACS3 also introduces new capabilities designed for the single-cell era. It can now read single-cell ATAC-seq fragment files directly, keeping cell barcodes and allowing researchers to call peaks on specific groups of cells, such as clusters from downstream analysis. This makes it possible to define cell-type specific accessible regions in a streamlined workflow. To further improve ATAC-seq analysis, MACS3 introduces hmmratic, a dedicated peak caller that uses a hidden Markov model to combine nucleosome-free and nucleosomal signals for more sensitive and precise detection of open chromatin regions. MACS3 also adds the ability to call genetic variants, e.g. SNVs and small indels, directly from functional genomics data, enabling studies of allele-specific accessibility and regulatory variation. These advances transform MACS3 from a general peak caller into a comprehensive toolkit for accessible-chromatin analysis, single-cell regulatory profiling, and regulatory variant discovery, helping researchers uncover the regulatory code of cell identity and state.

grassp: An integrated framework for GRaph-based Analysis of Spatial/Subcellular Proteomics

Max Frank

Tuesday, 15:45

Subcellular proteomics describes techniques that resolve proteome-wide organization in cells, revealing protein relocalizations that abundance-only assays miss. Protocols span gradient/fractionation profiling, organelle marker-based enrichment, and proximity labeling which all enrich local cell neighborhoods. Heterogeneous protocols and variability in dimensionality, quality, and noise profiles complicate analysis and cross-dataset integration. We introduce grassp (Graph-based Analysis of Spatial/Subcellular Proteomics), a Python framework designed to extract biological insights directly from raw mass spectrometry data. grassp is agnostic to the specific subcellular proteomics protocol and can be used to integrate multiple modalities such as proteins and lipids. At its core, grassp projects raw proteomic profiles onto a k-nearest neighbor graph, based on protein-protein distances. This graph-based representation forms the foundation for downstream analyses, including dimensionality reduction, unsupervised clustering, label transfer and relocalization detection. Because of the similarities of these analyses to single-cell multi-omics workflows, grassp builds on the robust infrastructure developed by the scverse community, allowing integration with single-cell analysis tools such as scanpy. Adopting the AnnData format provides a standardized and portable container for raw and processed data as well as analysis results. Leveraging this foundation, we build interactive visualization tools and a data warehouse for uniformly processed public datasets, allowing users to upload grassp-generated AnnData objects, perform quality control across multiple datasets, and explore or manually curate results through a web-based interface. We are excited to support a growing proteomics community in adapting scverse tools into their workflows. See our work at www.github.com/czbiohub-sf/grassp.

InterSCellar: Surface-Based Cell Interaction Volume Analysis in 3D Spatial Omics

Eunice Lee

Tuesday, 16:00

Background: Understanding cell-cell interactions is critical for uncovering disease mechanisms. Despite high-throughput imaging technologies shifting the spatial omics field into 3D space, current toolkits fail to capture biochemically interpretable neighborhoods and intercellular communication.

Objective: We present InterSCellar, a scverse-compatible Python package for surface-based cell neighborhood identification and intercellular volume computation in 3D spatially resolved datasets.

Methods: InterSCellar takes cell segmentations and raw expression profiles in de facto standard formats including SpatialData and OME-NGFF. The pipeline consists of: (1) surface-based detection of neighboring cell pairs within a user-defined distance, represented as a cell-neighborhood graph; (2) computation of intercellular spaces between neighbors as integrals of shortest Euclidean distances between surfaces; (3) calculation of intra/intercellular subvolume statistics. Bounding-box rendering, centroid-based pre-filtering, and parallelized multi-tasking optimize speed and scalability.

Results: We analyzed multiplexed CyCIF melanoma in-situ and invasive melanoma tissue samples to evaluate utility. For 11.3K cells distributed across a volume of 2754x5454x194 voxels, InterSCellar identified 26.7K+ neighboring cell pairs in <30 minutes then constructed intercellular volumes that accurately reflected known biomarkers by cell type, revealing stage-specific differences in interaction profiles at subcellular resolution. Graph-based representations further enabled instant expression comparisons between cell-cell pair types by aggregating subvolumes across whole images.

Conclusion: InterSCellar transforms high-dimensional spatial data into processed statistics for downstream biochemical analysis. Results can be visualized in Napari and Vitesse, enabling interactive exploration for data structures such as AnnData or SpatialData. This scalable tool extends the scverse ecosystem into 3D space, enabling spatial genomics, proteomics, and metabolomics analysis for therapeutic strategies.

****Would prefer oral presentation, but please consider for poster / flash talk otherwise.****

EasyVitesce: adding interactivity to Scverse single-cell and spatial biology plots

Eric Mörth

Tuesday, 16:15

EasyVitesce is a Python package that turns existing static Scanpy and SpatialData plots into interactive visualizations by virtue of adding a single line of Python code. The package uses Vitesce internally to render interactive plots, and abstracts away technical details involved with configuration of Vitesce. There are numerous benefits of interactive plotting in the context of single-cell data analysis. For instance, lasso interactions in scatterplots enable selection of cells of interest, which can be used for downstream analysis such as subclustering. In plots of spatial and imaging data, interactions such as zooming and panning enable smooth navigation from overview-to-detail. In all plots that display one biomarker (or a limited number of them) at a time, interactivity can enable users to easily select alternative biomarker(s) to view their expression patterns. EasyVitesce enables seamless transition between static and interactive plotting. The Python package leverages existing static plotting APIs available from the Scanpy and SpatialData-Plot packages so that users do not need to modify their plotting function calls. The resulting interactive plots can be viewed in computational notebook environments or their configurations can be exported for usage in other contexts such as web applications, greatly enhancing the utility of Scverse plotting APIs.

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Predictive modeling of molecular activity underlying physical cell-cell interactions

Tamjeed Azad
Poster 1

Interactions between cells are fundamental to tissue organization and function in health and disease. LIPSTIC (Labeling Immune Partnerships by SorTagging Intercellular Contacts) is a recently developed experimental technology that enables quantitative measurement of direct physical cellular interactions. LIPSTIC can be combined with single-cell RNA sequencing (scRNA-seq) for simultaneous high-throughput profiling of both cell-cell interaction intensities and intracellular molecular phenotypes at the single cell level. Here we present Gloss (Group Lasso on scRNA-seq), a novel predictive modeling framework for LIPSTIC+scRNA-seq data that systematically associates gene activity, measured via scRNA-seq, with cumulative cell-cell interaction intensities, measured by LIPSTIC. Using multiple publicly available datasets, curated ground truth annotations, and semi-simulated data, we demonstrate that Gloss achieves superior predictive performance compared to simpler baseline approaches. We apply Gloss to characterize molecular features of T cell interactions with myeloid cells during anti-Ctla4 immunotherapy in a mouse tumor model, and to describe interactions between different T cell subpopulations during viral infection. Gloss provides a broadly applicable computational framework for analyzing LIPSTIC+scRNA-seq data and prioritizing candidate genes and molecular pathways for future studies of cellular interactions.

Prediction of cellular morphology changes under perturbations with a transcriptome-guided diffusion model

Yimin Fan

Poster 2

Investigating cell morphology changes after perturbations using high-throughput image-based profiling is increasingly important for phenotypic drug discovery, including predicting mechanisms of action (MOA) and compound bioactivity. The vast space of chemical and genetic perturbations makes it impractical to explore all possibilities using conventional methods. Here we propose MorphDiff, a transcriptome-guided latent diffusion model that simulates high-fidelity cell morphological responses to perturbations. We demonstrate MorphDiff's effectiveness on three large-scale datasets, including two drug perturbation and one genetic perturbation dataset, covering thousands of perturbations. Extensive benchmarking shows MorphDiff accurately predicts cell morphological changes under unseen perturbations. Additionally, MorphDiff enhances MOA retrieval, achieving an accuracy comparable to ground-truth morphology and outperforming baseline methods by 16.9% and 8.0%, respectively. This work highlights MorphDiff's potential to accelerate phenotypic screening and improve MOA identification, making it a powerful tool in drug discovery.

Pycea: A Scverse Compatible Toolkit for Lineage Analysis

William Colgan

Poster 3

Lineage tracing technologies are rapidly advancing our ability to reconstruct cell histories in development, regeneration, and disease. However, standardized tools for analyzing lineage data within existing single-cell analysis frameworks remain limited. To address this gap, we developed Pycea, a Python package for lineage analysis and visualization that integrates seamlessly with the popular Scverse ecosystem. Pycea is built around TreeData, an extension of the AnnData format that adds support for storing phylogenetic trees. TreeData serves as a drop-in replacement for AnnData and remains fully compatible with Scanpy and other Scverse tools, enabling users to incorporate lineage information directly into standard single-cell workflows. Pycea provides a suite of scalable methods for visualizing lineage trees, identifying clades, comparing distances, inferring ancestral states, and computing phylogenetic statistics. To demonstrate these capabilities, we apply Pycea to a spatially resolved lineage tree of *C. elegans* development, illustrating how cell-intrinsic programs and extrinsic cues interact to define its deterministic development. By introducing a standard data format and a modular analysis toolkit, Pycea aims to accelerate the adoption of lineage-informed approaches across the single-cell community. The software is open source and available at <https://github.com/YosefLab/pycea>.

Joint inference of cellular phenotypes and clonotypic diversity of T-cells

Nicholas Ceglia
Poster 4

Single-cell T-cell receptor (TCR) repertoire sequencing enables simultaneous profiling of gene expression and TCR sequences, providing a co-registered measure of T cell phenotypic diversity and clonal dynamics. To model the relationship between genotype and phenotype in T cell biology, we introduce TCRi — a hierarchical Bayesian framework for inferring the joint probability distribution of clonotype and phenotype with paired transcriptomic and TCR data from the same cell. TCRi enables sharing information across patients and covariates to learn clonotype-specific phenotype distributions, while using a variational autoencoder (SCVI) to model count-based transcriptional expression. We applied TCRi to four large-scale real-world datasets comprising 304,550 CD8 T cells from patients with cancer treated with anti-PD-(L)1 immune checkpoint blockade. Leveraging information-theoretic metrics derived from the inferred joint TCRi distribution, we quantified the dependence between clonotype and phenotype. Notably, mutual information emerged as a robust discriminator of therapeutic response, underscoring TCRi's potential to reveal biologically meaningful links between T-cell clonal identity and transcriptional state in the context of immunotherapy.

Validating Omics Datasets Prior to Download: Streamlining Pre-Download Assessment of GEO DataSets

Maycon Marção
Poster 5

Efficient evaluation of publicly available omics datasets—like single-cell RNA-seq, spatial transcriptomics, and DNA methylation arrays—remains a bottleneck for researchers. Large repositories such as the Gene Expression Omnibus (GEO) offer thousands of datasets, but the lack of standardized, accessible metadata often forces users to download and manually inspect files, wasting valuable time and resources. Many datasets show inconsistency—missing gene names, incomplete spatial details, or fragmented files—highlighting the need for greater upfront transparency. We present an ongoing project aimed at transforming this assessment step with a semi-automated tool, enabling dataset validation prior to download. Our prototype, GEOmicScoutData, aggregates key metadata variables (including file format, gene annotation coverage, and spatial coordinate inclusion) from GEO DataSets using custom web scraping routines. This tool compiles relevant metadata into summary tables and visual aids, such as bar plots and filterable tables, helping researchers efficiently screen and select datasets. Compared to solutions like Polly and GEO2R, which focus on downstream analysis, GEOmicScoutData provides critical dataset-level overviews at the earliest stage of exploration, minimizing wasted effort and improving data handling workflows. Although web scraping presents technical challenges, our initial results demonstrate feasibility and value. We will present our system’s design, key findings, and plans to further automate pre-download evaluation for GEO and other omics repositories.

PEAKQC: Periodicity Evaluation in scATAC-seq data for quality assessment

Jan Detleffsen

Poster 6

Assessing chromatin accessibility at single-cell resolution using ATAC-seq presents unique challenges, particularly in quality control (QC). However, selecting high-quality cells for downstream analysis is crucial to ensure that biological insights are derived from meaningful data rather than noise. While robust QC standards exist for bulk ATAC-seq, single-cell ATAC-seq suffers from data sparsity, high noise levels, and the absence of universally accepted QC metrics and thresholds. One critical limitation is the inadequate characterization of fragment length distribution (FLD) at the single-cell level, while it is commonly used for bulk data. Here ratio-based approaches lack the effectiveness to capture the full potential FLD patterns can provide about data quality. We introduce PEAKQC, a Python-based tool that refines single-cell ATAC-seq quality assessment by leveraging a wavelet-based convolution approach to analyze FLD. Compared to conventional QC metrics, PEAKQC provides a more consistent and linear quality evaluation, enabling the identification and removal of low-quality cells affected by biological and technical artifacts. By enhancing QC filtering, PEAKQC improves downstream analyses, including clustering performance and cell-type identification, especially when used alongside existing metrics such as the fraction of fragments in peaks and total fragment counts. PEAKQC is integrated into the *sever* ecosystem, ensuring seamless compatibility with *scanpy*. The tool is openly available on Github and through the Python Package Index (PyPi). By addressing key deficiencies in current QC strategies, PEAKQC establishes FLD as a critical quality proxy at the single-cell level, offering a scalable and robust solution for the field.

Mechanistic inference of cell state transitions via Boolean GRNs from single-cell data

Roger Casals Franch
Poster 7

Understanding the regulatory logic behind cell fate decisions remains a major challenge. Cell state transitions are orchestrated by transcription factors (TFs) that regulate gene expression. These TFs form interconnected Gene Regulatory Networks (GRNs) whose structure determines stable cell identities and the paths between them. Reconstructing such networks from single-cell data is essential to uncover how gene regulation governs differentiation. However, current GRN inference methods, including multi-omic approaches, often fail to capture the causal mechanisms that drive transitions between cell states. These methods typically recover TF-target associations or marker genes but do not predict attractor states corresponding to known cell types, limiting mechanistic interpretability.

To overcome these limitations, we are developing a novel approach that identifies the most differentially active TFs across cell types and uses them to build an initial GRN. This network is then optimized using an objective function based on a deterministic Boolean solver, which identifies attractor states that match known cell types. To incorporate biological plausibility, our method minimizes asynchronous path lengths from initial to attractor states, assuming that differentiation follows the simplest trajectories. This allows us to pinpoint TF interactions that directly govern state changes.

We evaluated our method in recovering signed regulatory interactions, achieving an AUC of 0.782, outperforming GRNBoost (0.421) and GENIE3 (0.451). Furthermore, our framework can capture the regulatory logic whether TFs act cooperatively or independently.

This strategy aims to uncover regulatory circuits that drive cell state transitions, particularly enabling predictive modeling of differentiation under perturbed conditions.

From Millions of Cells to Molecular Mechanisms Through Scalable Image-Based Perturbation Screens

Noorsher Ahmed
Poster 8

Systematic mapping of genotype-to-phenotype relationships requires scalable methods that integrate spatial proteomics with genome-wide genetic perturbations. To address this challenge, we developed DECODE, an automated platform for pooled optical CRISPR screening (OPS) capable of executing genome-scale perturbations alongside high-dimensional spatial proteomics assays. DECODE leverages an AI-driven agent to systematically generate unsupervised target lists, design CRISPR guide libraries, and orchestrate robotic automation for iterative imaging and chemistry workflows. To process multi-million cell imaging experiments into "perturb-seq" style cell-by-feature tensors, we developed DECODE-im, which utilizes dask-parallelized and GPU-accelerated computer vision and image processing. Integrated within DECODE-im is the capability to generate large unsupervised feature embeddings per single cell across all protein channels using SubCell, a foundation model trained on the Human Protein Atlas. By combining the capabilities of the DECODE platform with the existing suite of scverse tools, we elucidate the genetic circuitry governing macrophage biology in unprecedented detail through the highest-dimensional imaging-based pooled screen of macrophages to date.

High-resolution human fetal retina and retinal organoids atlas to guide retinal cell differentiation and replacement efforts

Emil Kriukov
Poster 9

The complexity of the mammalian retina arises from the unique combination of extrinsic and intrinsic factors in its development. Here, we propose to investigate human retinal development and establish an approach the questions of cell fate specification, cell class and cell type heterogeneity, and regulation of cell trajectory by intrinsic and extrinsic factors at any given point in development. We establish a human fetal retinal and human retinal organoids atlas by integrating currently available scRNAseq datasets into a resource to create a high-resolution map of retinal development. We used advanced integrated transcriptomics methods, including pseudotime with the cell fate trajectory reconstruction, potency and trajectory density methods to deconvolute the transcriptional signal. We demonstrate the application of pseudotime approaches in reconstruction of retinal ganglion cells development to unravel the heterogeneity behind cell maturation. We discover hundreds of de novo development-oriented targets from both cell-intrinsic and cell-extrinsic perspectives of retina and retinal ganglion cells maturation. We further decipher maturation in vivo and in vitro to study how maturation states contribute to forming the environment sufficient retina development in general and between the conditions. We demonstrate the comparison and output of 12 organoids differentiation protocol with such metrics as cell commitment, similarity to native retina, development timepoint correlation, and transplantation probability prediction. We integrate and publish two new tools for the data analysis: a human fetal retinal atlas and human retinal organoids atlas. Both the atlases can be used as a community tool for analyzing, reference mapping and annotation of the sequencing data.

scVIP: personalized modeling of single-cell transcriptomics for developmental and disease phenotypes

Hsin-Yu Lai

Poster 10

Single-cell RNA sequencing (scRNA-seq) enables the discovery of cellular heterogeneity, but linking individual-level phenotypes to cellular states remains a major challenge. We introduce scVIP (scVI personalized), a generative framework that integrates scRNA-seq data with phenotypic markers using a variational autoencoder combined with a cell type-aware multi-instance learning (MIL) architecture. Unlike previous approaches such as multiMIL, which directly predict phenotypes from cell populations, scVIP leverages MIL to learn individual (donor)-level embeddings within the generative model. These embeddings jointly capture genetic, cellular, and phenotypic variation, enabling interpretable predictions of development and disease progression. Applied to mouse visual cortex data, scVIP accurately predicted developmental age and recovered known cellular trajectories and gene expression programs. In Alzheimer's disease datasets (ROSMAP and SEA-AD), scVIP outperformed existing methods in predicting neuropathology measures and staging, while harmonizing across cohorts with differing phenotype definitions. When trained on non-neuronal data, scVIP identified disease-associated microglia and astrocyte populations, with attention weights highlighting relevant biological pathways. By providing a scalable and interpretable framework for integrating molecular and phenotypic variation, scVIP lays the groundwork for subject-specific analyses of development and disease.

Expanding scRNA-seq analysis with literature-enriched single-cell embeddings using pre-trained language models

Sonia Maria Krißmer
Poster 11

The use of pre-trained language models for single-cell analysis is gaining increasing attention, yet most existing approaches are designed and evaluated with a focus on common data-intrinsic analysis workflows such as cell type annotation. Here, we demonstrate how lightweight encoder-only language models can enrich scRNA-seq pipelines for aligning data-derived and literature-derived representations within a shared embedding space and adding contextual layers such as functionality, disease, and temporal dynamics.

We construct purely text-based, structurally parallel training datasets from biomedical literature and scRNA-seq data, where single cells are represented as sentences derived from ranked gene expression and available meta-data. This allows for training lightweight encoder-only models jointly on both, text-based expression data and literature sources with a contrastive loss to learn an aligned representation. This strategy yields the integration of additional layers of knowledge from biomedical literature for obtaining literature-enriched single-cell embeddings, that can be used for a context-aware complementary downstream analysis based on relationships not previously captured in the single-cell dataset.

Analyzing the contextualized single-cell embeddings, we showcase downstream applications including cell type functionality annotation, identification of functional programs and related marker genes, and exploration of disease relationships. In an exemplary application to developmental data, incorporating temporal metadata enables capturing transitions consistent with lineage trajectories. The framework also provides systematic embedding-based evaluation strategies for language models in single-cell biology, including similarity tests with labels and metadata, annotation accuracy, and trajectory alignment to quantify how well models capture biologically meaningful relationships. More generally, we demonstrate how lightweight encoder-only language models can be used to add contextual layers of information to single-cell datasets and explore applications building on the literature-enriched cell embeddings.

Creating a Donor Foundation Model for the AI Virtual Cell

Kuan Pang

Poster 12

Learning universal representations of the complex multicellular signatures required for diagnosis and understanding of disease is a critical component of the AI Virtual Cell (AIVC). Here, we present PULSAR (Patient Understanding Leverage Single-cell universal Representation), a new multi-cellular foundation model that learns donor-level representations from single-cell RNA profiles. Trained in a full self-supervision on 36 million cells from over 6,800 samples, PULSAR integrates molecular, cellular, and multicellular information to generate robust donor-level embeddings that capture diverse biological signatures such as disease state, age, and immune function.

PULSAR demonstrates exceptional performance across diverse applications. In zero-shot settings, PULSAR consistently outperforms existing methods across disease classification with blood samples, circulating protein biomarker prediction, and cytokine perturbation response tasks. For example, donor embeddings accurately decode plasma protein concentrations across hundreds of immune and metabolic pathways, while reference mapping across large-scale PBMC atlases enables robust disease classification across autoimmune, infectious, and other inflammatory conditions. In addition, PULSAR predicts individualized cytokine responses, providing a framework for in silico immune perturbation and personalized therapeutic evaluation.

Beyond predictive accuracy, PULSAR remains interpretable: its attention mechanisms highlight cell types and gene programs most relevant to specific diseases and even stratify patients by severity, offering mechanistic insight into immune dysregulation. Together, these features establish PULSAR as a scalable, interpretable, and generalizable foundation model for clinical translation, capable of transforming large-scale single-cell atlases into actionable patient-level insights for diagnostics, prognosis, and therapeutic guidance.

Enhancing Spatial Transcriptomics Segmentation with Imaging-Based Approaches to Capture Irregular Cell Morphologies

Emily Thomas
Poster 13

Spatial transcriptomics (ST) provides powerful insights into tissue organization, yet segmentation remains a critical bottleneck. Poly-lobated and irregularly shaped cells, which play key biological roles, are often missed or misrepresented, even when boundary stains are incorporated. Existing segmentation strategies rely on nuclei- or radius-based priors or transcript localization, each of which simplifies cell shape and neglects the complexity of irregular morphologies. Imaging data, by contrast, provides higher fidelity cell boundaries and thus offers an opportunity to improve segmentation in ST. We propose an approach to integrate imaging-based segmentation directly into an existing ST segmentation. Beyond cell-level boundaries, constructing spatial graphs is central to downstream interpretation. Most studies adopt Delaunay triangulation to model cell-cell proximity, but this method struggles with irregularly shaped cells, often missing close connections. We highlight the potential of Voronoi-based graph construction, which can better account for insertion artefacts and the spatial relationships of non-uniform cells. To bridge computational and clinical practice, we additionally incorporate functions to exclude tissue regions that are not spatially preserved derived from pathologist-friendly GUIs, ensuring higher-quality downstream analysis. Imaging-derived features can also be overlaid to enrich ST data with morphological context. By combining imaging-informed segmentation, flexible region selection, and biologically faithful graph construction, our framework improves representation of short-range cellular interactions in ST datasets. We are implementing this methodology as a Python package, providing user-friendly tools for integrating imaging and ST workflows to better model tissue microenvironments and the roles of irregularly shaped cells in health and disease.

Spatial ligand–receptor prioritization with ancestry weighting and variance screening

Oona Shigeno Risse-Adams
Poster 14

Atlas-scale spatial assays now profile millions of cells, yet many ligand–receptor (LR) analyses depend on cluster labels, miss interface-level and regionally varying signaling, or do not scale to whole-slide data. We present a module of the Spatial Atlas of Human Anatomy (SAHA) that prioritizes LR channels from single-cell and spot-based data using two complementary readouts: a per-cell hotspot score for visualization and a neighborhood-weighted, edge-level spatial score for global ranking that scales to entire sections. In colorectal cancer, the framework recovers canonical epithelial–stromal axes (EREG→EGFR, COL1A1→ITGA11), separates niche-specific from density-driven signals with type-to-type summaries, and replicates LR rankings across Visium HD and whole-transcriptome CosMx SMI using a single LR catalog, supporting cross-platform validation. We augment scoring with population-aware priors summarizing external evidence for ancestry-differential risk, integrating cross-ancestry GWAS and cis-eQTL heterogeneity with regulatory context and transcriptomic replication. We further outline a noisy-label-tolerant gradient-boosted meta-model that learns non-linear mappings from these assembled priors to calibrated gene probabilities, producing interpretable weights for LR partners without requiring donor genotypes. By coupling spatial scoring with ancestry-aware prioritization, we build an interpretable, scalable tool for multimodal cell atlases. This design enables rapid hypothesis generation about microenvironmental mechanisms underlying population disparities while remaining platform-agnostic and extensible to additional priors as resources evolve.

Robust Single cell Meta-analysis Enables Molecular Dissection of Gastric Cancer Subtypes

Junha Cha
Poster 15

Gastric cancer (GC) is a heterogeneous disease with histopathological subtypes—intestinal, diffuse, and mixed—that differ in prognosis, metastasis, and treatment response. While morphological distinctions are recognized, the molecular and cellular programs remain poorly defined. Integrating single-cell datasets is particularly challenging at the molecular level due to technical variation. We developed an integrative framework combining in-house and public multi-omics datasets, including bulk sequencing, scRNA-seq, spatial transcriptomics, and proteomics. scRNA-seq data from 208 samples (118 patients) were harmonized and validated in The Cancer Genome Atlas (64 diffuse, 180 intestinal, 19 mixed). To address heterogeneity, we built a computational pipeline incorporating generalized linear models, non-negative matrix factorization for meta-program identification, sample-level pseudo-aggregation, and network-based gene expression analysis. This enabled gene-wise dissection of malignant transcriptional programs and systematic quantification of cell state dynamics across integrated cohorts. We identified distinct tumor-intrinsic programs across subtypes. Intestinal and mixed tumors preferentially expressed immune-associated programs, while MSI tumors exhibited elevated cell-cycle signaling compared to MSS. In the tumor microenvironment, diffuse tumors showed reduced B-cell differentiation but increased dendritic cell abundance, whereas intestinal tumors were enriched for T-cell subtypes, including regulatory T cells with immunosuppressive potential. TIGIT expression in CD8 and regulatory T cells was confirmed in an independent 142-patient cohort by multiplex immunofluorescence. Fibroblast-driven interactions dominated diffuse tumors, while intestinal and mixed tumors displayed immune-mediated “hot tumor” phenotypes. Computational integration of multi-cohort single-cell data revealed malignant and microenvironmental programs distinguishing GC subtypes. Our robust bioinformatics framework highlights subtype-specific vulnerabilities and enables therapeutic insights through gene-wise biology from public single-cell cohorts.

SpaTranslator: A Unified Framework for Spatial Multi-Omics Cross-Modality Generation

Sheng Mao
Poster 16

Recent advances in spatial omics technologies have enabled the simultaneous profiling of multiple molecular modalities within tissue slices, offering unprecedented insight into cellular microenvironments. However, paired spatial multi-omics datasets remain limited due to high measurement costs and data sparsity. To address these challenges, we introduce, a unified generative framework for cross-modality translation in spatial omics. SpaTranslator integrates Graph Convolutional Networks (GCNs) to alleviate spatial sparsity, applies Mutual Nearest Neighbor (MNN) loss for batch effect correction, and leverages a Variational Autoencoder (VAE) and Generative Adversarial Network (GAN) architecture for cross-modal generation. We comprehensively evaluate SpaTranslator in diverse spatial omics scenarios, particularly focusing on cross-modal generation challenges such as spatial RNA-to-ATAC and RNA-to-protein translation. Our approach outperforms state-of-the-art methods in clustering accuracy and biological coherence, and provides biologically meaningful insights through marker gene recovery and motif enrichment analysis. SpaTranslator establishes a flexible and effective foundation for spatial multi-omics integration and supports the broader vision of constructing virtual cells through AI-based data synthesis.

SingleRust: A High-Performance Toolkit for Single-Cell Data Analysis at Scale

Ian Ferenc Diks
Poster 17

Single-cell RNA sequencing studies increasingly generate datasets exceeding 10 million cells, surpassing the memory capacity of standard analytical tools on typical institutional infrastructure. Here we introduce SingleRust, a computational framework that addresses these constraints through systematic algorithmic optimizations and systems-level design. Key improvements include sparse masked principal component analysis that reduces memory footprint while preserving biological signal, lock-free parallel implementations for differential expression testing, and adaptive k-nearest neighbor algorithms that automatically select optimal data structures based on dataset size. These optimizations achieve 2.3-25.4-fold performance improvements and 1.1-4.0-fold memory reduction compared to Scanpy, enabling routine analysis of 30 million cells on our representative test system with 512 GB RAM. Comprehensive validation confirms numerical equivalence with established methods while maintaining biological interpretation fidelity. SingleRust maintains full compatibility with the AnnData ecosystem while providing researchers immediate access to population-scale analyses on existing infrastructure, addressing a critical bottleneck in single-cell genomics workflows.

(All of the code is available on: <https://github.com/SingleRust>)

Segger: Fast and accurate cell segmentation of imaging-based spatial transcriptomics data

Andrew Moorman
Poster 18

Accurate assignment of transcripts to their cells of origin is a central challenge in imaging-based spatial transcriptomics, limiting nearly all downstream analyses. Conventional cell segmentation methods often over- or under-segment, misassign transcripts, rely on manual intervention, and lack sensitivity and scalability.

We present segger, a graph neural network that reformulates segmentation as a transcript-to-cell link prediction problem using a heterogeneous graph of transcripts and cells. Segger can also incorporate single-cell RNA-seq data to refine assignments.

Across multiple Xenium benchmarks, segger achieves higher sensitivity and specificity than existing approaches while reducing computational cost by orders of magnitude. The open-source software is user-friendly, well-documented (https://elihei2.github.io/segger_dev/), and integrates seamlessly into current workflows. By minimizing manual intervention and enabling atlas-scale applications, segger provides a scalable and accurate solution for spatial transcriptomics analysis

SCENIC+ analysis identifies intermediate CD8 T cell states and dysregulated developmental timing in type 1 diabetes

Arpit Mishra
Poster 19

Single-cell multi-omics offers detailed insight into immune regulation by capturing current transcriptional snapshots (scRNA-seq) alongside chromatin accessibility states that reveal regulatory priming (scATAC-seq). These complementary layers are connected through transcription factor activity, and SCENIC+ eRegulon analysis integrates all three components to provide comprehensive views of cellular regulatory landscapes. We applied this approach to profile naïve and memory CD8 T cells from healthy individuals, people at risk for type 1 diabetes (T1D), and T1D patients, identifying regulatory programs governing naïve-to-memory transitions through region and gene-based AUC signatures.

We discovered a naïve cluster (C4) exhibiting mixed regulon usage characteristic of both naïve and memory states, an intermediate population that single-omics approaches would miss. Subclustering of cluster C4 combined with Milo differential abundance testing revealed distinct disease-specific enrichment patterns. Healthy individuals showed enrichment of negative regulatory control regulons (HIVEP1, RELB, ARID4B, ZEB1), while disease-associated samples exhibited elevated expression of inflammatory regulons (ETS1, CREM1, RFX3) coupled with increased homing and activation markers including SELL, ITGA4, and CXCR3.

Palantir pseudotime analysis consistently demonstrated that disease-associated and at-risk cells occupy earlier pseudotime positions compared to healthy controls, indicating a less differentiated, premature regulatory state. These cells displayed stem-like characteristics similar to those reported by Gearty et al., with ADT analysis confirming high CD62L, ITGA4, and CXCR3 expression patterns that mirror their findings. Disease and risk groups consistently exhibited lower pseudotime values than the healthy cohort across multiple trajectory analyses.

This work maps the regulatory landscape connecting naïve and memory CD8 T cell states and identifies disease-specific regulatory programs that may drive autoimmune progression in type 1 diabetes. Our discovery that T1D-associated cells exhibit premature developmental timing and a less-differentiated, stem-like phenotype provides new insights into T cell dysfunction in autoimmunity. These key observations will be further validated in future work using bulk RNA and ATAC sequencing.

A spatiotemporal virtual cell foundation model for development and diseases

Siyu He

Poster 20

Cell identity and fate are governed by the dynamic information flow encoded in cellular transcriptomes across time and space. While current high-throughput molecular profiling enables characterization of cell identity, limitations in spatiotemporal resolution restrict the ability to model various dynamic processes. Advances in generative AI combined with massive single-cell datasets across species and engineered tissues make it possible to build foundation models for virtual cells. Here we present Devo, a spatiotemporal, diffusion-based virtual cell foundation model that enables prediction and generation of cellular processes in healthy, diseased, and therapeutic contexts. Devo is pretrained on large-scale cross-species embryogenesis data, and fine-tuned with engineered tissue datasets, spanning normal, diseased, drug-treated, and spatial temporal data. Devo enables the prediction of cell transcriptomic states under a flexible design of conditions, including development, disease progression, and drug perturbation. Devo has shown the capability to generalize in human development across various cell types, and effectively indicated the developmental defect of the neurological diseases. When applied onto tumor cells, Devo serves as a powerful platform for drug screening. Furthermore, we applied Devo on tabula muris data and showed a powerful platform as a predicting spatial mouse aging atlas. Together, our results establish Devo as a foundation cell model for in silico exploration of molecular landscapes, and provides a powerful platform for precision and regenerative medicine.

CASCADE: Context-Aware Multi-Scale Modeling of Human Diseases

Valentina Giunchiglia
Poster 21

Human diseases entail interconnected changes that span molecular perturbations, cellular dysregulation, tissue remodeling, and ultimately clinical phenotypes. Capturing these multi-scale processes requires not only examining each level independently but also tracing how changes propagate across domains. Achieving such a holistic view demands studying cells within their biological context—defined by the tissue, disease, treatment, and cell type—and disentangling donor-specific features from those that truly reflect disease mechanisms.

Yet current modeling approaches remain limited. Cell-centric models extract knowledge from healthy cells in isolation, constraining their ability to learn patient-level features and thus fall short in predicting clinically relevant phenotypes. Patient-centric models frequently fail to recover detailed cellular context, generalize across donors, or provide mechanistic explanations grounded in genes and pathways.

We present CASCADE, a multi-scale, biologically interpretable single-cell model that bridges these gaps. CASCADE captures disease- and cell-relevant features across diverse conditions, generalizes across donors, and yields biologically grounded explanations. We demonstrate these capabilities by achieving superior performance in both cellular and clinical prediction tasks, outperforming leading single-cell foundation models. In addition, we investigate how CASCADE’s contextual learning capabilities enable further biological interpretation, designing a method to probe cell-type specific signatures in clinical phenotype prediction as well as gene-level and pathway-level importance derived from an explainability module. CASCADE demonstrates translational impact: it identifies biologically meaningful features across diseases, predicts therapeutic targets, and transfers knowledge across experimental and biological systems (e.g., organoids to human). Together, these results establish CASCADE as a powerful framework for uncovering disease mechanisms and accelerating precision medicine.

proSCessoR: A unified multimodal pipeline for enhanced single-cell immune repertoire analysis in vaccine research

Constanza Marini Macouzet
Poster 22

Single-cell omics technologies have revolutionized immunological research by enabling comprehensive profiling of cellular heterogeneity across multiple molecular dimensions. However, integrated analysis of multimodal single-cell data—particularly the combination of RNA expression, cell surface proteins, sample multiplexing tags, and immune receptor sequences—remains challenging due to fragmented computational tools. Here, we present proSCessoR, a unified single-cell multi-omics pipeline specifically designed for immunological research that seamlessly integrates RNA, cell surface protein (CSP), hashtag oligonucleotide (HTO), and B/T cell receptor (BCR/TCR) analysis within a single framework. We benchmarked proSCessoR against leading alternatives using simulated datasets and three human experimental datasets from COVID-19 and HIV vaccine studies. proSCessoR demonstrated superior performance in paired BCR capture compared to MiXCR and comparable performance to Cell Ranger, while achieving significantly more accurate somatic hypermutation quantification aligned with established biological principles. Notably, proSCessoR successfully recovered a higher number of rare B cell signatures in 82% of the test cases, a critical capability for studying broadly neutralizing antibody responses. The pipeline's integrated analytical framework enabled identification of clonal expansion and B cell differentiation trajectories following vaccination that would be difficult to detect using fragmented approaches. proSCessoR addresses critical gaps in single-cell immune repertoire analysis by providing a comprehensive, user-friendly platform that bridges computational analysis with experimental validation by generating high-confidence BCR sequences that enable direct functional applications, including monoclonal antibody generation. Available as both an R package and through the Cirro platform, proSCessoR represents a significant advancement in immune repertoire profiling for vaccine research.

PerTurbo: A scalable Bayesian framework for whole-transcriptome differential expression analysis and optimal design of single-cell CRISPR screens

Logan Blaine

Poster 23

Pooled single-cell CRISPR screens have revolutionized functional genomics by enabling large-scale perturbation studies of the genome. Yet current computational analysis approaches suffer from critical limitations. Existing fast statistical approaches produce many false positives, but more robust alternatives are markedly slower, making accurate transcriptome-wide testing for differentially expressed genes challenging.

To address these challenges, we developed PerTurbo: a scalable, fully Bayesian analysis framework for large-scale Perturb-seq experiments. PerTurbo uses GPU-accelerated stochastic variational inference, implemented with scvi-tools and Pyro, to efficiently analyze perturbation effects in datasets comprising millions of cells and thousands of perturbations. Crucially, unlike previous statistical tools, it also models CRISPR guide efficiency as a source of technical variability, allowing more accurate effect size estimation. We benchmarked PerTurbo using simulated datasets, where it performed competitively with existing methods in accurately identifying differentially expressed genes. To assess real-world performance, we reanalyzed an enhancer-mapping screen and found that PerTurbo identified similar enhancer-gene pairs compared with existing robust methods, while running $>10\times$ faster. This enabled us to exhaustively test for trans effects on distal genes, revealing enhancers of GATA1 and LMO2 whose repression affected hundreds of genes. To underscore PerTurbo's scalability, we next reanalyzed a genome-wide Perturb-seq screen with ~ 2 million cells in under two hours on a single machine, replicated key findings from the original analysis, and detected effects for "weak" perturbations overlooked in the previous analysis. Finally, we developed a simulation-based power analysis tool to help determine optimal sequencing depth and cell requirements for a given budget and experimental design.

Meningioma cell reprogramming and microenvironment interactions underlie brain invasion

Ayush Aggarwal
Poster 24

Meningioma brain invasion encumbers surgical resection and increases the risk of tumor recurrence, but the molecular mechanisms underlying this process are poorly understood. To identify molecular and cellular features of brain-invasive meningiomas, we (1) analyzed bulk RNA sequencing data from 199 meningiomas, including 33 brain-invasive tumors, (2) analyzed patient-matched single-cell RNA sequencing data of spatially mapped meningioma samples from the tumor core or brain-tumor interface (BTI), and (3) performed spatial transcriptomic sequencing of brain-invasive meningioma samples. Multiplexed immunofluorescence (IF) was used to validate bioinformatic spatial expression patterns. Functional interactions between meningioma cells and neurons were studied in meningioma/neuron co-cultures using confocal microscopy, multi-electrode array recordings, and live cell calcium imaging. Transcriptomic analyses showed conserved enrichment of TGM2, S100A11, ZYX, and PDGFRA at the BTI across bulk, single-cell, and spatial RNA sequencing datasets. The expression of these genes at the BTI was confirmed using multiplexed IF, and single-cell bioinformatic and imaging analyses further demonstrated enrichment of macrophages at the BTI. Co-culture assays showed neuronal hyperexcitability and increased proliferation of meningioma cells, suggesting functional communication between meningioma cells and the tumor microenvironment may contribute to meningioma growth in cases with brain invasion. In conclusion, meningioma brain invasion is defined by molecular remodeling of tumor cells and functional interactions within the tumor microenvironment.

scGETUVI: Bi-modal cell embeddings of gene expression and splicing isoforms illuminate cellular differentiation

Friedrich-Maximilian Weberling
Poster 25

Single-cell transcriptomic analysis of cellular differentiation is usually based on biological variations in protein-coding gene expression levels used to derive a unimodal Gaussian cell embedding using variational autoencoders. This only provides a coarse-grained view on the cell state dynamics because it ignores the process of splicing where one gene give rise to an entire family of proteins. Investigating spliced gene isoforms through transcript usage analysis will shed light on an important driver of evolution and biological complexity by providing a fine-grained view on the cell state. Both gene expression and transcript usage contribute to observed phenotypes and should be considered holistically. We introduce the deep generative model “single-cell gene expression and transcript usage variational inference” (scGETUVI) that based on a bi-modal mixture of experts variational autoencoder learns shared and private cell embeddings of gene expression and transcript usage for cell clustering and cell type prediction. The shared cell embeddings are modelled as a bi-modal Gaussian mixture model reflecting the patterns common in both gene expression and transcript usage whereas the unimodal private cell embeddings capture features specific to the gene expression or splicing profile in a cell. The clustering of the shared and private cell embeddings reveals discrimination between cell types that are otherwise hard to distinguish by solely considering gene expressions without information from spliced isoforms. The model also learns the importance of gene expression and splicing profiles per cell. The embeddings can then subsequently be used for cell type prediction of unseen gene expression and splicing profiles.

SCALPEL for isoform quantification at single-cell resolution

Franz Arnold AKE

Poster 26

Single-cell RNA sequencing (scRNA-seq) facilitates the study of transcriptome diversity in individual cells. Yet, many existing methods lack sensitivity and accuracy. Here we introduce SCALPEL, a Nextflow-based tool to quantify and characterize transcript isoforms from standard 3' scRNA-seq data. Using synthetic data, SCALPEL demonstrates higher sensitivity and specificity compared to other tools. In real datasets, SCALPEL predictions have a high agreement with other tools and can be experimentally validated. The use of SCALPEL on real datasets reveals novel cell populations undetectable using single-cell gene expression data, confirms known 3' UTR length changes during cell differentiation, and identifies cell-type specific miRNA signatures regulating isoform expression. Additionally, we show that SCALPEL improves isoform quantification using paired long- and short-read scRNA-seq data. Overall, SCALPEL expands the current scRNA-seq toolkit to explore post-transcriptional gene regulation across species, tissues, and technologies, advancing our understanding of gene regulatory mechanisms at the single-cell level.

SPOTTER: Bulk RNA-Seq Atlas Guided Annotation of Tumor Single-Cell Transcriptomes

Timmy Wen
Poster 27

Single-cell RNA sequencing (scRNA-seq) enables high-resolution profiling of cellular heterogeneity, revealing novel transcriptomic states. However, many existing generative and foundational single-cell models are trained predominantly on non-neoplastic data, limiting their accuracy in cancer cell annotation. We present SPOTTER (Seed-guided Prediction Of Tumor Transcriptomes with Ensemble Recognition), a framework that integrates bulk RNA sequencing (bulk RNA-seq) cancer atlases with single-cell data. SPOTTER first classifies individual cells using an ensemble neural network classifier OTTER (Oncologic TranscripTome Expression Recognition), trained on the hierarchical RACCOON (Resolution-Adaptive Coarse-to-fine Clusters OptimizatiON) cancer atlas spanning over 15,000 pediatric and adult cancer samples. High-confidence seed labels are selected using a Gaussian mixture model (GMM) and Gini impurity-based filtering of OTTER scores to exclude low-quality cells with uncertain predictions. These labels are then propagated through scANVI (single-cell Annotation using Variational Inference) to achieve per-cell classifications. Across ten diverse pediatric and adult single-cell and single-nucleus cancer datasets, SPOTTER reliably assigned malignant cells to their expected tumor classes. In Ewing sarcoma samples with matched bulk RNA-seq and single-nucleus RNA-seq (snRNA-seq), SPOTTER recapitulated bulk RNA-seq-defined subtypes at single-cell resolution, distinguishing one subtype enriched for neuronal programs, including SYT1 and SOX6, and another with increased EWS-FLI1 fusion activity and elevated JAK1 signaling—consistent with subtypes previously identified by OTTER and RACCOON in bulk RNA-seq. By integrating bulk and single-cell analyses, SPOTTER enables characterization of tumor heterogeneity and supports identification of subtype-specific markers to reveal critical insights into the transcriptomic profile of a cancer.

PyTorch Lightning on Ray for Distributed Training

Dung (Ivy) Tran

Poster 28

Computational biology increasingly relies on deep learning for analyzing complex biological datasets, yet the technical complexity of distributed training often prevents biologists from focusing on scientific discovery. We introduce PROTOplast, a package leveraging PyTorch Lightning and Anyscale Ray, to provide scalable machine learning for molecular data. This abstract highlights practical training and integration, while a separate abstract addresses technical architecture. To launch a run, users only need to define a model and prepare metadata, with resources management such as GPUs or workers specified through the Ray cluster configuration. RayTrainRunner takes care of scaling across large datasets by automatically managing GPUs and launching a cluster if needed. DistributedAnnDataset streams batches directly from large AnnData files, automatically sharding work across GPUs and workers, while allowing users to flexibly select or extend data fields as needed. Metadata such as gene names or cell identifiers are added through a callback, giving models the context required for training. Using this workflow, we trained three classification models, including SIMS, Autoencoder, and the scVI classifier training plan, as well as two perturbation models, EmbedSum and PerturbMean from State, all on Plate 3 and 12 (the smallest and largest file) of the Tahoe 100-M single cell dataset. One-epoch runs showed efficient runtimes: 16 to 60 minutes for classification models and approximately 42 to 120 minutes for perturbation models. Our results show PROTOplast can reduce time-to-experimentation by 70–80% in large-scale single-cell ML training, empowering biologists to iterate rapidly on hypotheses and advance scientific discovery without being constrained by infrastructure.

Mapping the ISR Landscape in Cognitive Disorders via single-cell multi-omics

Kristof Torkenczy
Poster 29

Persistent activation of the integrated stress response (ISR) is a major driver of cognitive decline in both neurodevelopmental and neurodegenerative disorders. Using a new mouse model (Ppp1r15b-R658C mice) that mimics the persistent ISR activation and cognitive decline observed in humans, we generated the first single-cell ISR atlas of the brain. By integrating single-cell RNA-seq and single-cell ATAC-seq with proteomics, we discovered that distinct brain cell types respond differently to persistent ISR activation and elicit cell-type-specific ISR programs. Interestingly, chromatin accessibility analyses revealed that the ISR downstream factor ATF4 is a key ISR effector in GABAergic neurons, while AP-1 (JUNB) is implicated in glutamatergic neurons. More importantly, selective deletion of ATF4 in GABAergic neurons—but not in glutamatergic neurons—impacts ISR-mediated cognitive decline in Ppp1r15b-R658C mice, demonstrating that different neuronal subtypes rely on unique ISR downstream effectors to regulate mnemonic processes. Furthermore, we defined a comprehensive molecular signature of persistent ISR activation, which we showed could serve as a biomarker for cognitive dysfunction across neurodevelopmental, neurodegenerative disorders and normal aging. This multi-omic framework provides a key platform for exploring and validating new scientific hypotheses, significantly advancing our understanding of ISR-related brain disorders.

scSketch: Interactive Directional Sketching for Single-Cell Analysis in Jupyter

Askar Temirbek
Poster 30

We present scSketch, a new interactive visualization and analysis tool for the scverse ecosystem that unifies programmatic workflows with exploratory sketch-based interfaces. Built on the GPU-accelerated jupyter-scatter widget and the anywidget framework, scSketch introduces interactive directional sketching—a novel approach to trajectory-like analysis where users define directions of interest directly in the embedding space (e.g., UMAP). This enables rapid identification of genes and pathways associated with continuous cellular transitions, complementing pseudotime and differential expression methods. scSketch integrates seamlessly with AnnData/Scanpy, allowing users to load existing objects, visualize embeddings, color cells by metadata, and export selected subsets. Beyond visualization, it couples sketch-based selections with statistical modules: online False Discovery Rate for error control during interactive analysis, integration with pathway resources such as Reactome, and support for negative-control simulations (via scDesign3) for robust inference. These capabilities transform exploratory interactions into reproducible analyses that can be embedded directly within Jupyter notebooks. We demonstrate scSketch using single-cell RNA-seq data from human cytomegalovirus (HCMV) infection of oral keratinocytes, where interactive sketching revealed immune and metabolic pathways modulated along infection gradients. More broadly, scSketch is applicable to any multimodal modality supported in AnnData except for spatial single-cell datasets (which are not yet supported). By extending the scverse ecosystem with real-time, error-checked directional analysis, scSketch lowers the barrier for rigorous hypothesis generation, facilitates communication between computational and experimental researchers, and opens new possibilities for interactive exploration of high-dimensional single-cell data. Project is hosted at <https://github.com/colabobio/scsketch> and an initial public release is forthcoming.

Unsupervised Deep Learning of DNA Methylation Aging Dynamics with Variational and Contrastive Autoencoders

Falak Pabari
Poster 31

DNA methylation clocks are powerful biomarkers of aging, but existing approaches predominantly rely on supervised regression or classification of CpG sites, which can obscure continuous or rare epigenetic patterns. Popular methods such as elastic net regression (Horvath, Hannum) and deep supervised CNNs (e.g., ResNet18 NoiseClock) achieve strong age prediction, yet their reliance on fixed labels limits the discovery of transitional states and unexpected biological trajectories. We present an unsupervised framework based on variational and contrastive autoencoders to model methylation–age dynamics at scale. Our dataset comprises over $\sim 470,000$ CpG-specific methylation trajectories. Initial VAE experiments with KL annealing demonstrated stable training, a significant reduction in reconstruction loss, and smoother manifolds with clearer inter-cluster separation than PCA and t-SNE baselines. These embeddings capture strong local coherence, though global organization remains diffuse. To address this, we are developing a hybrid framework that integrates contrastive learning with a variational autoencoder, using domain aware augmentations (curve jitter, scaling, noise) to enforce neighborhood structure in the latent space. Early experiments with the contrastive VAE are in progress, while current analyses overlay shape based traits (kurtosis, skew, linearity) to test whether embeddings reflect biologically meaningful variation. These findings suggest that unsupervised deep learning can complement supervised clocks by uncovering hidden methylation dynamics, providing a promising direction for identifying novel biomarkers beyond the scope of supervised clocks.

MultiTox: A Sequence-Based Stacked Ensemble Model for Multiclass Protein Toxin Classification

Harshika Sharma

Poster 32

Understanding the structural and functional diversity of toxin proteins is critical for elucidating macromolecular behavior, mechanistic variability, and structure-driven bioactivity. Traditional approaches have primarily focused on binary toxicity prediction, offering limited resolution into distinct modes of action of toxins. Here, we present MultiTox, an ensemble stacking framework for the classification of toxin proteins based on their molecular mode of action: neurotoxins, cytotoxins, hemotoxins, and enterotoxins. We curated a comprehensive dataset of 24,756 proteins (20,361 toxins and 4,395 non-toxins) and extracted high-dimensional ESM-2 embeddings that encode evolutionary, structural, and biochemical features. The two-tier stacking framework integrates LGBM, MLP, ET, KNN, and QDA as base classifiers and XGBoost as a meta classifier. MultiTox achieved an overall accuracy of 91.07%, an F1-score of 90.73%, and a Matthews Correlation Coefficient (MCC) of 91.61%. Class-wise accuracies were 93.75% (neurotoxins), 87.79% (cytotoxins), 98.80% (hemotoxins), 97.02% (enterotoxins), and 95.83% (toxins vs. non-toxins). SHAP-based interpretation and correlation with known physicochemical descriptors revealed class-specific features linked to biologically meaningful patterns in structural motifs, hydrophobicity, and solvent accessibility. Functional annotations using InterProScan, clusters of orthologs, and secretion signal analysis identified toxin class-specific signatures related to folding, localization, and host interactions. We deployed a public web server (<https://cosylab.iiitd.edu.in/multitox/>) for real-time and batch-mode predictions. MultiTox provides a scalable and biologically interpretable framework for protein classification, bridging sequence data with functional insights.

SpatialView-AI: Integrative, Interactive, and AI-Enhanced Visualization for Multi-Sample Spatial Transcriptomics

Chitrasen Mohanty
Poster 33

Spatial transcriptomics (ST) technologies are transforming biomedical research by enabling spatially resolved gene expression profiling, providing unprecedented insights into tissue organization and cellular interactions. With ST, investigators can address complex biological questions, such as how microenvironments differ across conditions or how disease progression alters tissue architecture. Tackling these questions requires multi-sample experiments, which depend on appropriate computational data preprocessing pipelines and advanced specialized analyses such as deconvolution. However, researchers often lack interactive visualization tools to explore integrated datasets critical for identifying spatial patterns and facilitating communication between analysts and investigators. To address this gap, we developed SpatialView, a scalable, secure application that runs locally on a simple HTML server, eliminating the need for uploading data to external servers or complex server management. Users can easily customize the application through a human-readable configuration file and optionally expose it for broader audience exploration. SpatialView supports dynamic grouping, statistical comparisons, and flexible visualizations, enabling intuitive exploration of multi-sample ST data while maintaining privacy. Seamless integration with the scVerse ecosystem allows easy export of processed data to SpatialView. We further extended SpatialView with advanced AI features, including context-aware queries and on-browser agentic workflows, to accelerate discovery and enhance collaborative interpretation. By bridging computational analyses and biological insights, SpatialView-AI provides a robust framework for interactive and collaborative exploration of spatial transcriptomics, empowering researchers to transform integrated data into refined hypotheses and actionable insights efficiently.

Effect of densification on CPU vs on GPU

Worapol Boontanonda

Poster 34

Single-cell RNA sequencing (scRNA-seq) datasets present unique computational challenges due to their extreme sparsity ($>90\%$ zero values) and massive scale, often containing millions of cells and tens of thousands of genes. Traditional data loading approaches either consume excessive memory through premature densification or create GPU utilization bottlenecks due to sparse tensor operations. This work introduces a specialized PyTorch DataLoader optimized for scRNA-seq data that addresses these memory and performance constraints through strategic GPU-based densification. The proposed DataLoader implements a multi-stage pipeline that maintains data in compressed sparse formats during CPU-based operations, including sampling, shuffling, and initial batch assembly. Data is transferred to GPU memory in compact sparse representations, where we perform efficient on-device densification just prior to model forward passes. This approach minimizes CPU-GPU data transfer overhead while maximizing GPU memory utilization and computational throughput. Key innovations include intelligent prefetching of subsequent batches during GPU computation, minimizing access to storage, and memory-mapped storage integration for seamless handling of datasets exceeding RAM capacity. The system incorporates dynamic load balancing across multiple GPUs and supports common scRNA-seq preprocessing operations including normalization and feature selection directly on GPU. Benchmarks on datasets ranging from 100K to 10M cells demonstrate 3-5x improvements in training throughput compared to standard approaches, while reducing peak memory usage by 40-60% on the host machine. The implementation maintains full compatibility with PyTorch's distributed training ecosystem, enabling scalable deep learning workflows for single-cell analysis applications including cell type classification, trajectory inference, and generative modeling.

Disentangling cell-intrinsic and cell-extrinsic factors underlying evolution

Alexander Starr

Poster 35

A long-standing question in biology is the extent to which cells function autonomously as opposed to requiring interactions with other cells or environmental factors. Here, we develop a framework to use interspecies chimeras to precisely decompose evolutionary divergence in any cellular trait into cell-intrinsic and cell-extrinsic components. Applying this framework to thousands of gene expression levels in reciprocal rat-mouse chimeras, we found that most divergence is cell intrinsic, though extrinsic factors also play an integral role. For example, cell-extrinsic regulation of a transcription factor can propagate to its target genes, leading to cell-type-specific extrinsic regulation of both their mRNA and their protein levels. We also show that imprinted genes are dramatically misexpressed in chimeras, suggesting a mismatch between rapidly evolving intrinsic and extrinsic imprinting mechanisms. Overall, our conceptual framework opens up new avenues to investigate the mechanistic basis of the evolution, development, and regulation of myriad cellular traits in any multicellular organism.

CytoVerse: A Browser-Native Platform for Scalable, Privacy-Preserving Single-Cell Transcriptomic Analysis and Reference Mapping

Jesus Gonzalez Ferrer

Poster 36

Mapping single-cell transcriptomic datasets to curated references has become a cornerstone of annotation and interpretation in single-cell RNA-seq. Existing web-based platforms such as Azimuth, SIMS-Web, MapMyCells, and ArchMap have broadened access to reference-guided analysis but remain constrained by server-side architectures that impose limits on dataset size, restrict interactivity, and raise concerns around data privacy. These bottlenecks are increasingly problematic as atlas-scale resources now encompass tens of millions of cells and as perturbation-based studies generate complex, out-of-distribution states absent from static training references.

We present CytoVerse, a fully browser-native platform for single-cell RNA-seq analysis and reference mapping. Built on WebAssembly (WASM), CytoVerse executes deep learning-based foundation models, nearest neighbor search, and cell type annotation directly within the browser, eliminating the need for data uploads or backend compute. This client-side architecture ensures real-time interactivity, supports streaming analysis of arbitrarily large datasets, and preserves data privacy while reducing computational infrastructure costs. By integrating foundation models for cell state representation, CytoVerse enables cross-study comparison of diverse datasets without manual harmonization or retraining.

Masked BERT-like pretraining of single-cell foundation models does not improve Virtual Cells

Bence Géza Czakó

Poster 37

Single-cell foundation models, pre-trained on large-scale single-cell RNA-seq data with masked LLM-style objectives, have been proposed as building blocks for Virtual Cells. Despite their popularity, benchmarks indicate that even trivial baselines often outperform them in downstream tasks. To evaluate the effectiveness of pretraining, we trained a transformer on 30M single cells from CELLxGENE, masking gene expression values. The model reached 0.76 Pearson correlation between predicted and observed masked genes. We then fine-tuned it on different post-perturbation RNA-seq prediction tasks, like (i) perturbation-exclusive (unseen perturbations, Replogle 2022), (ii) cell line-exclusive (unseen cell lines, McFarland 2020), and (iii) gene-exclusive (unseen mRNAs). Across these tasks, pretraining offered no advantage over training from scratch, except in the gene-exclusive setting where it conferred a modest benefit. Additionally, foundation model derived gene embeddings showed weaker performance than Gene Ontology when used as features in baseline machine learning models. Ablation studies showed that reducing transformer depth from 12 to 1 layers, varying masking ratios, decreasing gene context size and decreasing train data size to 4M single cells had little effect. These findings suggest that masked pretraining learns limited co-expression structure and does not transfer to perturbation prediction. Unlocking the potential of Virtual Cells may require richer pretraining objectives that leverage biological context such as temporal dynamics, spatial relationships, or multi-omics signals.

Harnessing single-cell multiomics to study the regulation of gene expression in T cells

Sarah Walker
Poster 38

T cells protect against pathogens and tumors and can differentiate into various functionally distinct subsets. While each subset exhibits a characteristic epigenomic and transcriptional profile, essential immune functions such as self-renewal, expansion, cytokine production, and cytotoxicity are common to multiple subsets and therefore must be programmed via shared regulatory mechanisms. To uncover these regulatory “archetypes”, we integrated a new single-cell multiomic (scATAC+RNA-seq) dataset for CD4 and CD8 T cells responding to acute and chronic viral infection with a comprehensive compendium of bulk ATAC-seq data. Because there is no state of the art pipeline for analyzing scATAC-seq data as there is for scRNA-seq analysis, we developed a new pipeline relying on standard scanpy tools to analyze our scATAC-seq data. Our analysis revealed that T cell identity is governed by a modular architecture, where distinct transcription factors drive the reuse of regulatory archetypes across T cell subsets and lineages. Our findings suggest a modular framework for regulating gene expression, where individual programs recur across multiple populations, and a single T cell state might exhibit multiple regulatory programs. Our framework for identifying shared regulatory programs, potentially driven by direct cell-cell interactions or shared microenvironments, can be extended to various other biological systems.

Discovery of a Linked Constellation of Gene Expression Revealed by Local Editing of Fibroblasts in Tumors

Nick Kuhn

Poster 39

Fibroblasts play critical roles in regulating cellular relationships during tissue homeostasis, immunity, and tumor biology at multiple sites. However, tools to perturb fibroblasts at just one site in vivo are limited, restricting our understanding of how these cellular relationships develop on a local level. We optimized local gene editing of fibroblasts in multiple mouse tumor models to investigate how locally restricted fibroblast perturbations affect the cellular tumor microenvironment (TME). By knocking out surface receptors *Osmr*, *Tgfr2*, or *Il1r1* on cancer-associated fibroblasts (CAFs), we uncover that TGFBR2 signaling loss uniquely induces the emergence of a *Col18a1hi* CAF cell state that is distinct from previously described fibroblast states and is associated with worse survival in human PDAC patients. Further application of a local as well as combinatorial gene knockout technology in CAFs reveals a circuit in which these *Col18a1hi* CAFs reshape the TME by recruiting Siglec-Fhi neutrophils via *Cxcl5* expression; and that the *Col18a1hi* CAF cell state is further dependent on TNFR1 and canonical Wnt signaling. Together, a fast, affordable, and modular engineering method is demonstrated, allowing discovery of a modified fibroblast identify, as well as the network details of a local inter-cellular circuitry in a tumor.

Hierarchical cross-entropy loss improves atlas-scale single-cell annotation models

Sebastiano Cultrera di Montesano

Poster 40

Accurately annotating cell types is essential for extracting biological insight from single-cell RNA-seq data. Although cell types are naturally organized into hierarchical ontologies, most computational models do not explicitly incorporate this structure into their training objectives. We introduce a hierarchical cross-entropy loss that aligns model objectives with biological structure. Applied to architectures ranging from linear models to transformers, this simple modification significantly improves out-of-distribution performance (12–15%) without added computational cost. Critically, we underscore the need to focus on new data generation that improves the connectivity among annotated cell types. Our work suggests this is likely to yield more generalizable algorithms than solely increasing model complexity.

Metagenome-assembled genomes enhance bacterial read decontamination and variant calling in oral samples

Zunu An
Poster 41

Whole genome sequencing (WGS) offers advantages over DNA chip-based genotyping, typically using blood-derived DNA. However, saliva and buccal samples—popular in direct-to-consumer tests—suffer reduced accuracy because of oral bacterial contamination. Decontamination strategies using decoy bacterial genomes yielded limited improvements, likely because they cover only a subset of oral bacteria with available isolate genomes. To overcome this, we developed a decontamination pipeline leveraging metagenome-assembled genomes (MAGs). Concordance analysis of variant calling between blood and matched oral samples confirmed the superiority of MAG-augmented decontamination over conventional methods relying mainly on isolate genomes. Although the underlying mechanism remains unclear, it particularly improves variant calls in GC-rich regions, recovering many likely pathogenic variants. Additionally, we demonstrate that certain bacterial genomic regions mimic human regions with clinically relevant variants, potentially confounding genotyping. These results highlight the need for MAG-based bacterial read decontamination to achieve accurate personal genotyping from non-invasive, self-collected oral samples.

Minibatch Loading using AnnData

Ilan Gold

Poster 42

Training smaller GPU-based models, like those from scvi-tools, on on-disk datasets is generally io-bound due to both the code doing the data loading and the file formats themselves. Thus a large chunk of the time spent training a model that repeatedly loads from disk is in reality file-io. To this end, we propose a new minibatch loading framework for the anndata file format. By using a chunked data-fetching strategy, zarr v3-based anndata files, and efficient in-memory data handling, we achieve large performance speedups over state of the art by addressing the two root cause of slow io speeds: the way bytes are stored on-disk and how they are read into memory. Furthermore, we are able to address these two bottlenecks while still building directly on the AnnData file format. This approach stands in contrast to other current approaches that might create new file formats or do not address data loading code bottlenecks directly. Thus the current approaches do not maximize io performance, and thereby cause slower fitting times for many methods. We also provide new out-of-core functionality for pre-shuffling the data and writing to the zarr v3 sharded format. Preliminary results suggest over 10X speedup over state of the art.

ProteoImager: an integrative platform for image coregistration, spatial analysis, and visualization for high-resolution spatial proteomics datasets

Zhenru Zhou

Poster 43

ProteoImager: an integrative platform for image coregistration, spatial analysis, and visualization for high-resolution spatial proteomics datasets

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The rapid growth of mass spectrometry-based spatial proteomics necessitates robust computational tools for integrating deep proteome profiling with spatial context, crucial for understanding cellular organization in health and disease. We introduce ProteoImager, an innovative Python Dash application offering a programming-free graphical user interface (GUI) for end-to-end spatial proteomics data analysis. Its unified Python backend and frontend are designed to provide a dynamic and responsive platform for data interaction and visualization. ProteoImager bridges critical technical gaps across whole-slide imaging (WSI), laser dissection (LCM), and LC-MS workflows. Key computational features include: • Accurate coordinate coregistration, transforming WSI coordinates to LCM coordinates using fiducial landmarks and determinant-based calculations for precise object extraction. • Flexible data input, accommodating proteomics data from tools like DIA-NN, Spectronaut, and FragPipe, alongside image processing outputs (e.g., GeoJSON from QuPath). • Comprehensive downstream analysis: quality control, normalization, dimension reduction (UMAP), and clustering (HDBSCAN). • Advanced spatial statistical analysis to identify differentially abundant and spatially variable proteins, with interactive visualizations overlaid on tissue sections or UMAP plots. ProteoImager directly addresses the conference themes of "Computational tools for spatial genomics and high-throughput imaging" and "Multimodal approaches to decode gene regulation and cell state dynamics," empowering intuitive exploration of complex spatial proteomic data.

CARIBOU: Computational AI Research Interface for Bioinformatics, Omics, and Unifying Agents

Varshini Vijay
Poster 44

The convergence of single-cell genomics and artificial intelligence offers opportunities but exacerbates challenges in reproducibility, accessibility, and scientific rigor. We introduce CARIBOU (the Computational AI Research Interface for Bioinformatics, Omics, and Unifying Agents), a framework that instantiates a new paradigm of human AI partnership. CARIBOU enables scientists to act as high-level strategists, directing autonomous agent teams to execute end-to-end single-cell analyses.

CARIBOU’s architecture orchestrates LLM-powered agents within secure, containerized environments (Docker or Apptainer/Singularity), ensuring every analysis is reproducible, inspectable, and HPC-compatible. CARIBOU introduces a Recursive Scientific Refinement loop, wherein agents receive structured feedback from metric evaluations (e.g., scib) and iteratively revise their strategies without human intervention.

CARIBOU agents operate within the scverse ecosystem, using scanpy for preprocessing and visualization, scvi-tools for deep generative modeling, and scanpy-rapids for GPU-accelerated workflows. Agents understand the data model behind anndata, enabling context-aware decisions that follow best practices and avoid common pitfalls in single-cell analysis.

In preliminary benchmarking, CARIBOU achieved a +15.2% improvement in total scib integration score over a baseline generic scvi-tools pipeline. When compared to a single-shot ChatGPT interaction (zero-shot prompt using only the dataset name and “perform single cell qc and run scVI”), CARIBOU succeeded where ChatGPT failed to complete execution, hallucinating invalid method calls such as `.get_latents()` instead of `.get_latent_representation()`.

By transforming analyses into auditable digital lab notebooks and embedding agent-based scientific reasoning into the workflow, CARIBOU sets a new standard for AI-native, reproducible, and adaptive research pipelines in computational biology.

A Fast, Distributed-Friendly DataModule for Set-to-Set Perturbation Training on AnnData

Tan Phan

Poster 45

Motivation Training perturbation models on large single-cell datasets is often bottlenecked by random I/O and inefficient batching. Conventional DataLoaders treat each cell independently, making multi-worker or distributed scaling I/O-bound instead of compute-bound.

Results We present a PyTorch Lightning DataModule for perturbation training that directly optimizes data access and batching. Our design relies on three principles: - Pre-ordered h5ad files by target gene and batch, enabling contiguous range reads instead of random access. - Range-based slicing of matrices allowing balanced partitioning across workers and distributed ranks while keeping disk reads sequential. - Set-to-set sampling, where each batch contains S perturbation cells for a single target gene paired with S matched control cells from the same cell type. Our region-based indexing further supports few-shot and zero-shot splits, maintaining I/O efficiency while allowing per-target train/val/test assignment.

Impact By coupling storage layout with batching semantics, our module reduces random I/O, improves throughput, and makes distributed perturbation training practical. Implemented as a Lightning DataModule, it integrates seamlessly into scverse workflows, lowering the barrier for scalable perturbation model development.

Fetal oocyte attrition: Quality control for the ovarian reserve?

Jay Zussman

Poster 46

The germ cell lineage gives rise to oocytes and sperm and is critical for transmitting genetic and epigenetic information across generations. In mammals, ovarian fertility depends on a finite ovarian reserve established in utero. In the mouse, the germline is initially specified as a small population of primordial germ cells (PGCs) that migrate asynchronously through the early embryo to reach the gonads. Gonadal PGCs undergo sex determination to oogonia and further expand in interconnected cysts to a peak population of ~25,000 before entering meiosis and differentiating into oocytes. Curiously, most nascent oocytes are immediately culled through fetal oocyte attrition (FOA), yielding ~8,000 oocytes in the initial ovarian reserve. Primordial germ cells, oogonia, and oocytes exhibit transcriptional heterogeneity in processes including epigenetic reprogramming, DNA damage repair, and meiotic entry timing. This variation, alongside the extensive reduction in oocyte population, has suggested that FOA functions as a selective quality control filter. However, whether attrition preferentially eliminates oocytes based on specific clonal features remains unclear. To address this, we harness single-cell lineage tracing using high-throughput inducible genetic barcoding to assess clonal transcriptional heterogeneity among early oocytes just before FOA. In addition, we apply Bayesian modelling to test whether probabilistic patterns of cyst breakdown can explain observed clonal dynamics in primordial follicle formation. These approaches provide a framework to distinguish stochastic from selective mechanisms of FOA and identify molecular features that may predict oocyte survival. This work will reveal developmental determinants of oocyte quality and guide efficiency improvements in human in vitro oogenesis protocols.

SLAF: Open-source cloud-native single-cell storage format for the virtual cell era

Pavan Ramkumar
Poster 47

Single-cell datasets have grown 2,000-fold in less than a decade, creating fundamental bottlenecks in traditional H5AD-based workflows. Current approaches require downloading entire datasets from object storage to local storage, and loading them into memory, making 100M-cell analyses impossible on commodity hardware.

SLAF (Sparse Lazy Array Format) addresses these limitations through a cloud-native format designed as a drop-in replacement for H5AD. Built on columnar storage with vector support and superior random access latency (Lance) and Rust-accelerated lazy query engines (Polars), SLAF enables lazy evaluation with familiar scanpy idioms such as numpy-like slicing, while supporting concurrent access from multiple users.

Key innovations include: (1) Query-in-place access eliminating data downloads, (2) Memory-efficient lazy operations enabling 100M-cell analysis on commodity compute, (3) pushdown optimization for complex filtering and aggregation, (4) Advanced thread-based dataloaders and tokenizers for foundation model training, achieving 20-100x throughput relative to TileDB-SOMA, AnnLoader, and AnnDataLoader.

We demonstrate SLAF's integration with existing scanpy pipelines, migration from H5AD workflows, and new capabilities for interactive exploration of atlas-scale datasets. SLAF is an MIT-licensed, open source package (github.com/slaf-project/slaf), with extensive documentation, and a growing suite of examples and benchmarks. SLAF represents a path forward for the scanpy ecosystem to handle the scale and complexity of the virtual cell era data.

The Bigger Picture: Single-cell Image Context Combined with Protein Language Models Improves Function-based Signals

Mohini Misra

Poster 48

A protein's function is intrinsically tied to its cellular location, influenced by its microenvironment and nearby interaction partners. However, the latent embeddings learned by protein language models like ESM2 are not designed to capture these critical nuances that arise with spatial context. SubCell, a recent vision transformer model, was trained on single-cell protein localization images to generate embeddings that represent protein spatial distributions. In this study, we integrate the embedding representations of ESM2 and SubCell, hypothesizing that the combination will produce a richer protein embedding informed by both sequence and single-cell context. First, we confirm that the image and sequence modalities contain distinct, orthogonal information, showcasing the utility of a multi-modal approach. To integrate the two into a multimodal embedding, we use an existing modality integration method that employs unsupervised learning. We show that this new embedding space is notably enriched for protein-protein interactions signals relative to the single-modal embeddings, to a more-than-additive degree. Additionally, the multimodal space better separates functionally divergent paralogs while maintaining high-similarity for functionally redundant paralogs — a balance that neither the sequence space nor image space can achieve alone. Finally, we demonstrate that location-specific biological processes are better enriched in the multimodal space, suggesting relevance to systems-level phenomena. Thus, a combined sequence-image embedding is a useful tool for downstream applications in functional annotation or pathway discovery. More broadly, this work underscores how single-cell images can inform generalized biological representations, moving us toward representations of molecules specific to a cell state.

Uncovering Spatial Cell-Cell Interaction Programs by Tailoring Cell Sequences for Transformer Neural Networks

Niklas Brunn
Poster 49

Emerging spatial transcriptomics techniques enable tracing cellular interactions in situ with unprecedented resolution. Yet, existing computational approaches lack the capacity to summarize interactions mediated through common cells within spatial niches and to prioritize candidate interactions from extensive ligand–receptor databases. These challenges are further compounded by increasing sparsity in spatial gene expression data.

To this end we present the Cell Sequences framework for utilizing transformer neural networks to uncover spatial cell-cell interaction programs at single-cell resolution. Cell sequences are built from random walks on a cell graph emulating how humans might intuitively trace cellular interactions within tissue. Each sequence links cells via a distance-aware guilt-by-association mechanism that borrows signal from transcriptionally similar neighbors to mitigate sparsity, and ligand-receptor co-expressions for known ligand-receptor interactions. To summarize interactions into programs, we employ a structured dimensionality reduction approach to inform communication edges in the cell graph for sequence construction. Further assuming that a receiver cell’s gene-expression profile is conditionally dependent on gene expression profiles of sender cells communicating with the receiver cell, we augment cell sequences with expression profiles of genes beyond those in interaction databases to promote transformer-based prioritization of interaction programs. This enhances interpretability of results visualization by leveraging learned attention weights.

In applications to human breast cancer spatial transcriptomics data, we showcase how cell sequences can facilitate either targeted interaction analysis for user-selected interactions or comprehensive exploration of all feasible communication mechanisms. Further, we demonstrate broad applicability to various spatial transcriptomics platforms, including single-cell-resolution technologies such as Xenium and Visium HD.

GenomicDataStream: A scalable interface between genomic data and analysis underneath R

Gabriel Hoffman

Poster 50

(Temp) Reading genomic data files (VCF, BCF, BGEN, PGEN, BED, H5AD, HDF5, DelayedArray) into R/Rcpp in chunks for analysis with Armadillo / Eigen / Rcpp libraries. Modern datasets are often too big to fit into memory, and many analyses operate on a small chunk features at a time. Yet in practice, many implementations require the whole dataset stored in memory. Others pair an analysis with a specific data format in way that the two components can't be separated for use in other applications. For example, regression analysis paired with genotype data from a VCF file. The GenomicDataStream interface separates data source, streaming chunks of features into a data matrix, downstream analysis. GenomicDataStream provides interfaces at both the C++ and R levels. The C++ interface prioritizes efficiency, while the R interface wraps the C++ backend for non-technical users.

Transcriptomic and Dynamic Modeling Defines a Self-Sustaining Inflammatory Loop in Post-COVID-19 Pulmonary Sequelae

Mohit Joshi
Poster 51

Abstract Background: A subset of individuals recovering from severe COVID-19 develop Post-Acute Sequelae of COVID-19 (PASC), marked by persistent pulmonary inflammation and dysfunction. The immunological mechanisms underpinning this chronic lung pathology remain incompletely understood. This study investigates immune signatures in bronchoalveolar T-cells of convalescent COVID-19 patients to elucidate drivers of PASC.

Methods: We re-analyzed publicly available single-cell RNA sequencing (scRNA-seq) data from bronchoalveolar lavage (BAL) T-cells of convalescent COVID-19 patients (n=5) and healthy controls (n=2). A computational pipeline using the Seurat R package was employed, including SCTransform normalization, anchor-based data integration, unsupervised clustering, and differential gene expression analysis. Functional enrichment was performed using DAVID and Reactome. A simplified three-component ordinary differential equation (ODE) model was constructed to simulate dynamic immune responses and disease progression.

Results: Transcriptomic analysis revealed a distinct separation between convalescent and control T-cells. Convalescent T-cells exhibited strong upregulation of pro-inflammatory genes, including cytotoxic effectors (e.g., GZMA, NKG7) and chemokines (CCL5). Clustering uncovered unique CD8⁺ T-cell and NK cell populations enriched only in the convalescent group. Functional enrichment showed activation of T-cell cytotoxicity, IFN- signaling, and leukocyte-mediated immunity. The ODE model demonstrated how a transient yet intense immune activation could evolve into a chronic inflammatory state, leading to irreversible lung damage consistent with PASC.

Conclusion: Our study indicates that chronic lung pathology in PASC stems from sustained, dysregulated T-cell activity rather than viral persistence. These insights reveal potential therapeutic targets aimed at modulating aberrant immune responses in post-COVID recovery.

Approachable and intuitive single-cell multiomic analysis with the BD Cellismo tool and Muon

Devon Jensen

Poster 52

We created a desktop application for analysis and visualization of single-cell multiomic data using a muon and python-based backend. Multiple modalities are supported, including RNA, PROT, and ATAC data, as well as cell hashing. Users can create eight different types of graphs and import or export data in multiple formats. Integrated tools for interpreting cell state include: visual quality control, batch correction, integration, doublet detection, cell type classification, clustering, differential analysis, and motif enrichment. Also included are a project organization system, with image gallery, and export to SVG and PNG. Our experience integrating scverse tools is discussed. <https://www.bdbiosciences.com/cellismo>

Pantheon: An Evolvable Distributed Multi-Agent System to Automate Single Cell and Spatial Genomics Data Analysis

Weize Xu

Poster 53

The rapid development of single-cell and spatial omics technologies has generated massive high-dimensional datasets. However, the complexity of computational analysis pipelines poses a major challenge for many biologists, constraining the pace of scientific discovery. To address this bottleneck, we developed Pantheon, an evolvable distributed multi-agent system designed to automate and intelligentize genomics analysis. To accommodate different user preferences, the Pantheon framework supports multiple interaction interfaces—command-line (CLI), web user interface, and Slack—allowing users to conduct complex end-to-end data analyses uniformly via natural language, enabling researchers without programming backgrounds to use it with ease. At its core is a distributed architecture that allows users to deploy and run analysis workflows in their local environments, ensuring the privacy and security of sensitive data, while also enabling efficient collaboration among specialized agents deployed across different compute nodes. To promote high extensibility and community collaboration, we also built the Pantheon Store, an open platform for sharing and reusing key components such as Agents, Toolsets, and Retrieval-Augmented Generation (RAG). More importantly, we introduce the Pantheon-Evolve framework, inspired by genetic algorithms, which leverages large language models not only to autonomously evolve analysis algorithms but also to optimize agents' behaviors and collaboration patterns, thereby surpassing existing methods and creating new analysis tools with superior performance.

Identification of Inflammation-Related Gene Signatures in Major Depressive Disorder Using Transcriptomic Workflows

Anny Silva Adri
Poster 54

Major depressive disorder (MDD) is a prevalent psychiatric illness with complex biological underpinnings and limited objective biomarkers. Advances in transcriptomics and computational workflows enable standardized evaluation of large-scale datasets, improving reproducibility and translational potential. In this study, we performed a meta-analysis of three independent public transcriptomic datasets (GSE32280, GSE98793, and GSE19738), totaling 348 individuals (210 MDD patients and 138 controls). Using a benchmarked pipeline combining limma for differential expression, MetaVolcanoR for integration, and logistic regression with stepwise feature selection, we identified 5,417 differentially expressed genes (DEGs). A minimal DEG signature achieved high discriminatory performance between groups ($AUC = 0.906$), and six genes (CEACAM6, CRISP3, FAM118A, FOLR3, LCN2, and OLFM4) remained consistently significant across datasets. Functional enrichment analysis revealed pathways related to interleukin signaling, neutrophil degranulation, and iron homeostasis, supporting the role of inflammatory processes in MDD. Notably, LCN2 and OLFM4 emerged as strong candidates for biomarker development, consistent with previous reports of their involvement in immune-related mechanisms. Beyond biological insights, this study highlights the value of standardized computational workflows for transcriptomic data integration, offering a reproducible framework that can be extended to other psychiatric and complex disorders. By connecting robust benchmarking strategies with molecular target identification, our approach contributes to the growing effort of applying computational biology to precision psychiatry and may inform future drug discovery pipelines.

Pancancer network analysis reveals key master regulators for cancer invasiveness

Mahesh Jethalia

Poster 55

Tumor invasiveness reflects diverse biological processes, including tumorigenesis, progression, and metastasis. To investigate the role of transcriptional regulators (TRs) underlying invasiveness, we conducted a systematic pan-cancer network-based assessment of master regulators (MRs) of cancer invasiveness.

Patients in The Cancer Genome Atlas (TCGA) were stratified into high (INV-H) and low (INV-L) invasiveness groups using consensus clustering based on a robust 24-gene invasiveness signature. We evaluated the prognostic relevance of invasiveness for overall survival (OS) across 32 cancers and devised a network-based framework to identify TRs acting as MRs unique to INV-H and INV-L phenotypes. Candidate MRs associated with INV-H and poor OS were further validated using independent datasets from the Prediction of Clinical Outcomes from the Genomic Profiles (PRECOG) repository.

Our analysis demonstrated that invasiveness was broadly associated with worse outcomes and significantly correlated with OS in 10 of 32 cancers. The network-based framework identified multiple invasiveness-associated MRs specific to INV-H and INV-L groups, including COL1A1, part of the original 24-gene signature, serving as a positive control. Pathway enrichment of INV-H-specific MRs revealed key biological programs, including epithelial-to-mesenchymal transition, TGF- signaling, Toll-like receptor regulation, cytokine and inflammatory responses, and selective chemokine receptor expression during T-cell polarization—pathways strongly linked to immune modulation and metastatic potential.

This pan-cancer study provides a comprehensive MR analysis of tumor invasiveness, highlighting conserved transcriptional drivers and pathways. These findings may inform the development of more precise therapeutic strategies by targeting invasiveness-associated regulators and their downstream signaling networks across multiple cancers.

Why do some joints get Rheumatoid Arthritis and not others?

Max Cowan

Poster 56

Rheumatoid Arthritis is an autoimmune condition characterised by its distinct pattern of polyarthriti-
tis and synovial inflammation, but also the accompanying systemic features such as autoantibody
production, cardiopulmonary disease, neurological complications, and skeletal disease. Ref 1 Al-
though the model of pathogenesis remains to be elucidated, fibroblast dysfunction is implicated as a
central feature. Key inflammatory fibroblast populations with discrete anatomical distributions have
been described and are of great interest as targetable populations, including 'universal' fibroblasts
(such as PI16+ populations) with the potential to restore homeostasis or promote resolution. Ref
2, 3 Across the interphalangeal joints of the hand, Rheumatoid Arthritis will affect the Proximal
Interphalangeal Joint ("PIP") while sparing the Distal Interphalangeal Joint ("DIP"), despite both
joints being similar synovial hinge joints. To inform our understanding of the pathogenesis of this
disease, this project seeks to construct a 3D reference atlas of the anatomical, cellular, and tran-
scriptomic differences across the two sites which drive this tropism.

Ref 1. McInnes, I.B. and Schett, G., 2011. The pathogenesis of rheumatoid arthritis. *New
England journal of medicine*, 365(23), pp.2205-2219 Ref 2. Croft, A.P., Campos, J., Jansen, K.,
Turner, J.D., Marshall, J., Attar, M., Savary, L., Wehmeyer, C., Naylor, A.J., Kemble, S. and
Begum, J., 2019. Distinct fibroblast subsets drive inflammation and damage in arthritis. *Nature*,
570(7760), pp.246-251. Ref 3. Wei, K., Korsunsky, I., Marshall, J.L., Gao, A., Watts, G.F., Major,
T., Croft, A.P., Watts, J., Blazar, P.E., Lange, J.K. and Thornhill, T.S., 2020. Notch signalling
drives synovial fibroblast identity and arthritis pathology. *Nature*, 582(7811), pp.259-26

COPPR: Copilot for Omics Processing and Profiling of RNA

Saurav Roy

Poster 57

Single-cell RNA sequencing workflows demand a seamless integration of alignment, quality control, statistical testing, and visualization, yet current approaches remain fragmented across heterogeneous tools and scripting environments. We introduce COPPR, a modular command-line framework implemented in Python that unifies standard RNA-seq alignment and post-processing with advanced downstream analytics for both bulk and single-cell data. The system builds on established tools such as STAR for alignment, samtools and Picard for sorting and duplicate marking, and RNA-SeQC for coverage assessment, but extends functionality through Python modules that handle statistical modeling and visualization. For bulk RNA-seq, COPPR supports differential expression via log fold-change estimation and Welch's t-test with false discovery rate correction, with optional execution of DESeq2 through R/Python integration. For single-cell data, it leverages the Scanpy/AnnData ecosystem to provide reproducible quality filtering, mitochondrial content analysis, normalization, highly variable gene selection, scaling, and dimensionality reduction using PCA, t-SNE, and UMAP. The tool outputs both data objects in interoperable formats and publication-ready figures such as volcano plots, clustering maps, and reduced-dimension embeddings. Designed with a Typer-based CLI, COPPR emphasizes interoperability with high-performance computing environments by supporting manifest-based STAR execution and strandedness-aware RNA-SeQC while maintaining compatibility with existing pipelines. By abstracting complex statistical and visualization tasks into reproducible command-line modules, COPPR reduces coding overhead while preserving analytical rigor, enabling rapid iteration from raw sequencing data to interpretable results. This framework offers an extensible, open-source foundation for computational and experimental laboratories engaged in single-cell transcriptomics.

GFP-2A-mRubyhTau CD1 murine model for tau uptake investigation

Daya Mena
Poster 58

Tauopathies are neurodegenerative diseases characterized by hyperphosphorylation and aggregation of the microtubule-associated protein tau. Glial dysfunction is increasingly appreciated as a disease phenotype that often precedes and contributes to tau pathology. Here, we employ a GFP-2A-mRubyhTau CD1 murine model to investigate tau uptake over a three-week incubation period. To assess the molecular consequences of tau spread, we apply spatially resolved transcriptomics to the affected hemisphere of injected mice. Using cell2location and Squidpy within the scverse ecosystem and CellChat, we deconvoluted cell type composition, identified spatially differential gene programs, and reconstructed cell-cell communication networks. We find that astrocytes and microglia in the spread hemisphere adopt pro-inflammatory states marked by elevated cytokine production and activation of Complement and Galectin signaling pathways. Communication analyses reveal disrupted astrocyte-neuron interactions, coinciding with upregulation of synapse regulation and remodeling pathways. Additionally, glial populations observe consistent downregulation of lipid transport and cholesterol metabolism programs in glial cells. Finally, CD45 expression suggests a contribution of both microglial activation and peripheral immune infiltration. Together, these findings highlight coordinated inflammatory, metabolic, and signaling reprogramming, as well as multi-lineage involvement during early tau spread, and showcase how scverse-enabled workflows enable the integration of deconvolution, spatial statistics, and network inference to dissect neurodegenerative disease mechanisms

A Multi-Platform Computational Benchmark for Detecting Somatic Mosaic Variants in Healthy and Alzheimer's Disease Brains

Jiaming Weng
Poster 59

Somatic mosaicism is increasingly implicated in neurological disorders, yet the accurate detection of low allele fraction mosaic variants (MVs) remains a significant challenge, with outcomes heavily dependent upon the choice of sequencing technology and computational pipeline. Our objective is to systematically benchmark the performance of leading-edge sequencing platforms and bioinformatic tools for identifying somatic MVs in the human brain. We have generated comprehensive sequencing data from the same healthy and Alzheimer's Disease (AD) brain tissues across multiple modalities, including Illumina short-read, Ultima PCR-free ppmseq, PacBio HiFi long-read, and Duplex sequencing, at depths from 30x to 300x. Additionally, we performed single-cell Primary Template-directed Amplification (PTA) sequencing to assess single-cell resolution and utilized ultra-deep (3000x) Massive Parallel Amplicon Sequencing (MPAS) as a ground truth set for clonal MVs. By applying a standardized suite of MV callers to this multi-platform dataset, we will quantify the sensitivity and accuracy of each technology and computational approach across a range of read depths. Our hypothesis is that each platform possesses unique biases in MV detection and that AD brains may exhibit an excessive burden of specific MV types or DNA damage signatures. This work will establish a critical framework for standardized evaluation in the field and guide the design of future studies investigating the role of somatic mosaicism in neurodegeneration.

QuantumCell

Margaret Alexander
Poster 60

QuantumCell addresses the critical challenge of accurately mapping and annotating cellular identities within complex tissues. It integrates advanced single-cell (sc) analysis methodologies, including variational inference, hierarchical dataset integration, and generative pre-trained transformers for gene expression modeling. In spatial omics, where cells are densely packed and boundaries overlap, traditional approaches often fail to resolve cell identities. QuantumCell overcomes these limitations by leveraging computational innovations inspired by multi-attention transformers and sequence-to-sequence vision–language encoder–decoder models. A key feature of QuantumCell is the generation of Metacell atlases from large-scale sc datasets hosted on CellxGene. By aggregating transcriptionally similar cells into robust Metacells, these condensed atlases provide high-confidence cell type and state references. When integrated with spatial omics data, Metacell atlases enable more accurate cell type assignment and contextual mapping within tissue architectures. A multimodal transformer fuses transcriptional and spatial features, yielding interpretable models of tissue composition, cellular neighborhoods, and molecular interactions. QuantumCell introduces a novel file format optimized for spatial omics, simplifying the integration of transcriptomic and proteomic measurements. This unified framework supports segmentation of complex datasets into biologically meaningful units, facilitating precise characterization of diverse cell populations. By linking curated Metacell atlases to spatial analyses, QuantumCell enhances resolution, consistency, and interpretability in tissue mapping. This approach promises to advance our understanding of spatial cellular dynamics in health and disease, enabling the discovery of novel biomarkers and therapeutic targets through a more faithful reconstruction of tissue organization at single-cell resolution.

SPixel: Integrating High-Resolution H&E Pathology with Visium HD, CosMx, and Xenium Spatial Transcriptomics for Enhanced Multi-Modal Analysis

Yutian Liu

Poster 61

Title: SPixel: Integrating High-Resolution H&E Pathology with Visium HD, CosMx, and Xenium Spatial Transcriptomics for Enhanced Multi-Modal Analysis Authors: Yutian Liu, Maycon Marcao, Maggie Alexander Felipe Pimenta Carcanholo, Felipe Dezem-Segato, Luke Zhang, Jasmine Plummer Affiliation: St. Jude Children's Research Hospital, Center of SpatialOmics

Spatial transcriptomics (ST) has emerged as a transformative technology for linking gene expression to tissue architecture. Platforms such as Visium HD (sequence-based) and CosMx/Xenium (image-based) enable high-resolution molecular profiling, yet their integration with conventional histopathology, particularly Hematoxylin and Eosin (H&E) staining, remains limited. Bridging these modalities enables linking molecular and morphological features in a unified spatial framework. A multi-modal integration pipeline was developed that overlays Visium HD, CosMx, and Xenium ST with high-resolution H&E pathology. We used pixel-based classification and GPU-accelerated clustering (RAPIDS) to segment H&E images into morphologically distinct regions. These morphological features were directly mapped to spatial gene expression profiles, enabling correlation between tissue architecture and molecular states. To allow for easier adoption, an interactive R Shiny application was built for dynamic exploration across modalities. Our approach demonstrated (1) robust alignment of pixel-based H&E features with both sequence- and image-based ST, (2) improved resolution in detecting cellular niches and tissue heterogeneity, and (3) interactive multi-layer visualization enabling real-time exploration of histological and molecular data. Importantly, the pipeline supports large-scale datasets with GPU acceleration, ensuring both analytical precision and scalability. By integrating high-resolution H&E pathology with Visium HD, CosMx, and Xenium, we present a reproducible, scalable framework for multi-modal spatial analysis. This work exemplifies how pixel-based histology can serve as a unifying layer to decode cell state dynamics, benchmark cross-platform datasets, and establish intelligent workflows for spatial genomics.

Improve Patient-Level Optimal Transport in Single-Cell Omics with Global Ground Metric Learning

Damin Kühn
Poster 62

Optimal transport (OT) is a powerful mathematical framework for comparing probability distributions with applications in patient-level analysis of single cell-omics. In this context, Patient samples are represented as distributions of single cells over the gene space. OT computes distances between these distributions to identify similarities and differences between patients which can be used for downstream-applications, such as classification, clustering and embeddings. Crucially, the effectiveness of OT is significantly influenced by the choice of the underlying ground metric (or cost). Traditionally, the ground metric has either been predefined (e.g., Euclidean or cosine), or learned in a supervised way, by utilizing labeled data to learn a suitable ground metric for enhanced task-specific performance. However, predefined metrics typically cannot account for the inherent structure and varying significance of different features in the data. Existing supervised ground metric learning methods often fail to generalize across multiple classes or are limited to distributions with shared supports. To address this issue, we introduce a novel approach for learning metrics for arbitrary distributions over a shared metric space. Our method provides a distance between points (cells) like a global metric, but requires only patient-level class labels (e.g., disease states) for training. The resulting learned global ground metric enables more accurate OT distances, significantly improving clustering and classification in scRNA datasets. This work was published at AISTATS2025 as “Global Ground Metric Learning with Applications to scRNA data”. Our Python package, with seamless AnnData integration, aims to equip the scverse community with a practical tool to enhance OT-based analyses.

Multimodal clustering reveals cellular heterogeneity and spatial organization in a single blastocyst

Lyda Yuliana Parra Forero

Poster 63

Understanding cellular heterogeneity is essential for deciphering early embryonic development. In this study, we applied three unsupervised clustering algorithms K-Means, DBSCAN, and Agglomerative Clustering to analyze the expression profiles of DAPI, GATA6, OCT4, and CDX2 in a single mouse blastocyst. Prior to clustering, Z-score normalization was performed to ensure comparability across markers. Each method revealed distinct aspects of the cellular landscape: K-Means provided a baseline segmentation into three groups; DBSCAN identified outlier cells potentially in transitional states; and Agglomerative Clustering offered a hierarchical view of cell affinities. Three-dimensional visualizations based on spatial coordinates enabled the interpretation of expression domains and subpopulation structure. While each method has limitations, their combined use allowed a more nuanced understanding of cell organization and marker co-expression. This integrative approach highlights the potential of clustering techniques to uncover biologically meaningful subpopulations and supports the need for expanded datasets to validate these findings across multiple embryos.

Single-Cell Perturbation Microscopy

Samantha Mutiti

Poster 64

Whole-genome perturbation screens paired with model-based image phenotyping are emerging as powerful approaches for mapping gene–phenotype relationships at scale. Innovations in optical pooled screening (OPS) and experimental automation now enable imaging-based perturbation experiments that approach the dimensionality of some of the largest single-cell RNA-seq atlases. However, the analysis of these datasets at the single cell level is constrained by a computational bottleneck: accurate and efficient single cell segmentation, registration, and phenotypic profiling across millions of cells per experiment. As OPS experiments continue to grow in scale, the need for performant and scalable computational infrastructure is critical to unlock the capability to perform 'single-cell microscopy' at the accelerated pace of genomic experiments.

We present Decode-im, a single-cell segmentation, registration, and analysis package as part of the Decode platform. Decode-im manages pooled screen datasets by (i) performing scalable cell segmentation, (ii) single cell registration across imaging rounds unconstrained by differences in resolution, magnification, or even imaging modality, (iii) assigning CRISPR perturbations at single-cell resolution, and (iv) embedding-based phenotyping of perturbed cells via integration with vision foundation models. We have achieved desired performance with GPU acceleration of image processing and dask-parallelized streaming of multi-terabyte datasets, allowing for whole datasets to be processed in a matter of hours. Finally, Decode-im organizes OPS datasets into scverse-compatible object formats, including AnnData and SpatialData, and provides OPS-specific analysis utilities accelerated with RAPIDS for handling multi-million-cell perturbation experiments. Together, these capabilities position Decode-im to accelerate the discovery of gene–phenotype relationships at single-cell resolution.

The Impacts of Dataset Imbalance on Single-cell Foundation Models

Izumi Ando

Poster 65

In single-cell RNA sequencing (scRNA-seq) research, a common analysis approach is to combine batches of data from different samples and experiments. Although routine, this integration step is not a trivial task because the batches being integrated often contain different cell-types and cell-type proportions. Foundation models pretrained with self-supervised techniques are a recent innovation that are developed to be an all-in-one solution for any scRNA-seq analysis step, including data integration. In this study, we aimed to address whether the presence of cell-type imbalance in the batches being integrated using foundation models affect biological signals in downstream analysis post-integration. Extending the pre-established Iniquitate benchmarking pipeline, we integrated four different datasets containing variable numbers of batches with non-foundation model integration tools (scVI and Harmony) and three foundation models (scGPT, Transcriptformer, Geneformer), in both zero-shot and fine-tuned settings. Using the integrated embeddings from each tool, we examined integration performance and the effects on downstream analyses, including cell-type classification, unsupervised clustering, and differential gene expression analysis. Our preliminary results show that batch-effects and dataset imbalance do affect downstream analysis in foundation models, and that the magnitude of these effects are cell-type specific. Understanding the limitations of scRNA-seq foundation models is essential to properly interpret the results yielded from them, as these tools become a core part of our research pipelines.

Neuroimmune Interactions in Colorectal Cancer Insights from Single-Cell RNA Sequencing Analysis

Débora Freitas
Poster 66

Colorectal cancer (CRC), the third most diagnosed cancer worldwide, poses a major public health burden and remains a leading cause of cancer-related mortality despite therapeutic advances. Neuroimmune interactions, involving complex communication between the nervous and immune systems, are increasingly recognized as pivotal in tumor biology. In CRC, such interactions may promote tumor growth, enable immune evasion, and facilitate metastasis. The nervous system shapes immune responses through neurotransmitters, cytokines, and neuropeptides, which can modulate immune cell activity, sustain tumor-associated inflammation, and enhance tumor survival. In this study, we integrated multiple public datasets using the SCVI (single-cell variational inference) framework to achieve robust harmonization of single-cell RNA sequencing (scRNAseq) data. This approach enabled a high-resolution dissection of gene expression patterns across heterogeneous immune cell populations. By applying comprehensive bioinformatics and statistical tools, we investigated neuroimmune interactions and identified specific pathways involved in CRC progression. Our analysis highlights that GABAergic and glutamatergic transmission, particularly in plasmablast/plasma B cells and myeloid cells within the tumor microenvironment, plays an important role in driving tumor development toward advanced stages, likely through modulation of immune activity and shaping of the tumor milieu. These findings provide detailed molecular insights into the neuroimmune mechanisms that influence CRC biology and may open avenues for the development of novel, targeted therapeutic strategies designed to disrupt these interactions. Ultimately, uncovering the contribution of neurotransmitter-driven immune modulation offers new opportunities for improving outcomes in CRC patients.

Cell type-specific gene network inference using foundation model embeddings and graph neural networks

Yurim Jung
Poster 67

Most existing approaches to study gene regulation rely on bulk-level pathway information and fail to capture regulation specific to each cell type. To address this limitation, we propose a framework that integrates single-cell foundation model embeddings with graph neural networks (GNNs) for cell type-specific gene network inference. For each cell type, we construct a context-specific graph where nodes represent genes and edges are labeled as positive or negative according to curated references. Gene features are derived from foundation model embeddings, aggregated by mean-pooling within each cell type. Independent GNNs are trained per cell type, with cross-validation designed to prevent data leakage across labeled and unlabeled edges. Our framework outputs probabilities of gene-gene interactions tailored to each cellular context, enabling systematic evaluation of cell type-specific networks. Initial results show encouraging trends across balanced accuracy, AUROC, AUPRC, and Brier score, suggesting that the model captures meaningful context-dependent patterns. Beyond labeled interactions, predictions for unlabeled gene pairs open the door to biological interpretation, with preliminary comparisons to external reference networks indicating reproducibility across immune cell types. Predicted networks further allow functional interpretation at multiple levels. GO enrichment analyses highlight pathways enriched in specific cell types, while hub gene analyses identify candidate regulators with potential biological relevance. Together, these results suggest that our framework provides a scalable strategy to uncover context-dependent gene networks, advancing single-cell inference and studies of immune regulation.

Single-Cell Transcriptomics Reveals Divergent Interferon Programs in Anti-TNF Responders and Non-Responders

Euijeong Sung
Poster 68

Tumor necrosis factor- (TNF-) inhibitors are widely used in the treatment of rheumatoid arthritis (RA), yet therapeutic responses remain highly heterogeneous, with a substantial proportion of patients failing to achieve clinical benefit. To elucidate the cellular and molecular determinants underlying differential responses, we performed single-cell RNA sequencing of peripheral blood mononuclear cells (PBMCs) obtained from RA patients before anti-TNF- therapy. Systematic analyses across immune subsets revealed distinct transcriptional programs associated with treatment outcome. Notably, interferon--related gene signatures and signaling pathways were significantly enriched in responders, whereas interferon--driven programs predominated in non-responders. Cell-type-specific analyses highlighted responder-enriched activation of interferon-stimulated genes within interferon classical monocytes, alongside perturbation-based modeling that confirmed their strong association with favorable clinical trajectories. In contrast, non-responders exhibited heightened interferon--linked activation in NK and T cell subsets, consistent with pro-inflammatory circuitries previously implicated in autoimmune pathogenesis. Collectively, our findings suggest that divergent interferon axis utilization, that type I interferon in responders versus type II interferon in non-responders, represents a critical immunological determinant of anti-TNF- efficacy in RA. These insights may inform predictive biomarker development and guide rational therapeutic stratification.

Genome-wide perturb-seq in primary human T cells maps regulatory networks controlling immune variation

Ronghui Zhu
Poster 69

CD4⁺ T cells orchestrate immune responses against infection and cancer, yet a comprehensive map of their gene regulatory networks has been lacking. Perturb-seq enables systematic mapping of such networks, but large-scale application in primary human T cells has been challenging. Here, we developed a scalable, probe-based Perturb-seq platform and performed genome-scale perturbations of all expressed genes across 22 million primary CD4⁺ T cells from four donors under three stimulation conditions. We demonstrate the utility of this reference map in three ways. First, we identify previously unrecognized regulators of the two important immune cytokines, IL-10 and IL-21. Second, we show that ex vivo perturbation signatures can model T cell states observed in population-scale atlases, revealing regulators of helper T cell polarization and T cell aging. Third, we leverage regulatory relationships to mechanistically explain gene-trait associations in immune-related diseases, nominating genes for uncharacterized immune functions. Our reference map provides a foundational resource for decoding immune variation in humans and reprogramming T cell states for next-generation precision therapies.

HTO-DND: A Robust, Noise-Aware Method for Demultiplexing Hashtag Data in Single-Cell Sequencing

Tobias Krause
Poster 70

Accurate assignment of cells to their samples in multiplexed single-cell sequencing is critical for reducing noise and maintaining statistical power in downstream analyses. However, conventional demultiplexing methods often ignore technical and biological variation or require manual tuning. This makes them unsuitable for processing samples uniformly. We present HTO-DND, a noise-aware demultiplexing method that normalizes differences in sample viability, technical handling, and library preparation while robustly assigning cells across heterogeneous datasets. HTO-DND was benchmarked on two in-house cell-line datasets with known ground truth, simulated variations, and an independent dataset. Across all conditions, HTO-DND achieved equal or higher sensitivity and precision compared with existing methods without parameter optimization. It is available as an open-source (https://github.com/sail-mskcc/hto_dnd/) Python library and CLI, available on PyPI and DockerHub. HTO-DND leverages scverse supported data structures and thus integrates seamlessly into existing workflows for robust hashtag demultiplexing, particularly useful for core facilities.

scProx: spatial E3–target matching

Rabia Khurshid

Poster 71

Background: We frame proximity-induced targeted protein degradation (TPD) as a modeling problem: given a tissue neighborhood, is a target degradable here, and which E3 should we recruit? We operationalize pre-design feasibility in situ, providing a context-aware spatial decision model—a Degradation Readiness Index (DRI; 0–1) and an E3 selection per spot—to focus subsequent degrader chemistry and design. By factorizing feasibility into interpretable axes (Target, recruiter/E3, UPS, NEDD8), scProx also advances pathway deconvolution of context drivers, not limited to E3. **Methods:** scProx is a scverse-native, target-agnostic workflow that computes DRI from scaled observables—Target expression, E3-module activity (CRBN/CUL4, VHL/CUL2, KEAP1/CUL3, DCAF15/16; extensible), and ubiquitin–proteasome/NEDD8 capacity—using AnnData/Scanpy and Squidpy on Visium/Census inputs. Each scProx Spatial Card standardizes a triptych—Target×E3 overlay, DRI (with contour), and DRI what-if—plus best-E3 and neighborhood enrichment. An optional AI layer (scvi-tools) yields cell-type–resolved DRI and uncertainty. **Results:** On public slides, scProx reveals context-dependent Target×E3 co-localization, recommends recruiters that vary by niche, and pinpoints neighborhoods with the largest DRI under ligase modulation. **Evaluation & Reproducibility:** Ablations versus an E3-only baseline, component sensitivity, cross-slide stability; Snakemake pipeline with pinned environments regenerates figures in minutes. **Availability:** Code will be released openly prior to the event; repository link will appear on the poster. **Relevance:** Beyond E3, the modular decision framework generalizes to pathway-aware context selection for other modalities (e.g., kinase pathway inhibitors, molecular glues, cell-engineering levers), linking spatial context to TPD feasibility, recruiter choice, and downstream chemistry.

Differential colocalization of cell-type pairs in single-cell spatial omics with DC-SPOMIC

Jake Chang
Poster 72

Advancements in single-cell spatial omics technologies allow unprecedented exploration of tissue microenvironments and the discovery of tissue-specific and cell type-specific, spatially-resolved molecular patterns associated with development, pathogenesis, and drug response. However, the process of pooling spatial estimates and findings across samples while hypothesis testing for meaningful differences between groups poses challenges under limited sample size and when the degree of heterogeneity within and between samples is high. To address this challenge, we introduce DC-SPOMIC, a tool for identifying differentially colocalized cell type pairs between groups of spatial omics samples. DC-SPOMIC uses local spatial statistics of each cell in a sample to calculate colocalization between pairs of cell types and spatial bootstrapping to estimate the degree of intra-sample variation of the colocalization estimate. The estimated intra-sample variances per sample are used in a random-effects meta-analysis model, which pools cell type pair colocalization estimates across samples while estimating the overall inter-sample heterogeneity. The random-effects output provides a pooled estimate of colocalization and its precision, which can be used for downstream analysis such as hypothesis testing of differential colocalization between groups of spatial omic samples. We demonstrate the robustness of colocalization estimation and pooling across samples using DC-SPOMIC on simulated data from spatial point process models. When used to analyze distinct histological groups of colorectal cancer samples generated by CODEX, DC-SPOMIC identifies seven statistically significant pairwise spatial colocalization differences amongst immune cell populations and network analysis revealed a subnetwork of four immune cell types exhibiting higher group colocalization. By quantifying both intra- and inter- sample variability, DC-SPOMIC offers potential to uncover significant spatial interactions in complex tissues and across heterogeneous datasets.

Identifying high resolution single-cell eQTLs with quasar and milo

Jeffrey Pullin

Poster 73

Identifying genetic variants that affect gene expression, expression quantitative trait loci (eQTLs), is a major focus of modern genomics. Recently, we developed quasar, a flexible and efficient C++ software program for eQTL mapping. Compared to existing methods, quasar implements a wider variety of statistical models, including both count-distribution and mixed effect models. In addition, quasar implements improved methods for estimating the negative binomial dispersion, previously used in the edgeR and DESeq2 packages, which we demonstrate improves Type 1 error rate control. However, quasar is only designed to work with bulk RNA-seq data, or pseudobulk data grouped into discrete cell-type clusters. The emergence of large disease-specific and population-scale single-cell RNA-seq datasets highlights a need for methods that can detect eQTLs at higher resolutions. We plan to extend quasar to be able to map such eQTLs by extending the milo framework for partitioning single-cell data into neighbourhoods. We will create a Python package miloQTL to identify neighbourhoods with the right properties for powerful eQTL testing, building on previous work to use neighbourhoods to identify differentially expressed genes and genetic variants associated with cell-state abundance. Gene expression aggregated by neighbourhood will be analysed by quasar to detect eQTLs. We expect these neighbourhood-defined eQTLs capture eQTLs at high resolution, for example in rare cell-types, along trajectories, and across continuous variations in cell-state. We are interested in collaborating with experienced scverse developers to ensure that miloQTL is a fully integrated component of the scverse ecosystem.

AnnSQL: a Python SQL-based package for fast large-scale single-cell genomics analysis using minimal computational resources

Kenny Pavan
Poster 74

The rapid growth of single-cell and single-nucleus RNA-seq datasets presents computational challenges for storage, querying, and analysis. We present AnnSQL, a Python package that combines an AnnData like schema with the embedded database DuckDB, enabling efficient SQL-based operations on large-scale single-cell data. AnnSQL supports both in-memory and on-disk workflows, returning results in familiar formats such as DataFrame, AnnData, or Parquet, and integrates into existing single-cell analysis pipelines.

Benchmarking demonstrates that AnnSQL achieves orders-of-magnitude improvements in filtering speed and preprocessing memory efficiency of large datasets over existing tools. For example, operations on datasets exceeding four million nuclei, often infeasible in other frameworks, can be completed on a modest laptop. This performance enables users to conduct complex queries and transformations without the need for specialized high-performance computing resources.

Since its release, AnnSQL has seen exciting adoption, with nearly 30k total downloads and consistent monthly usage, underscoring its utility for the single-cell community. By lowering computational barriers and enabling efficient exploration of atlas-scale datasets, AnnSQL expands access to single-cell analysis, making it a potential powerful addition users familiar with the scverse ecosystem.

Beyond the Usual Suspects: AI-Assisted Target Selection for Functional Genomics

Anton Thieme
Poster 75

High-throughput pooled perturbation screens, coupled with increasingly sophisticated analytical pipelines, are rapidly advancing functional genomics. Yet as these technologies scale, the bottleneck is shifting upstream to the design phase, where the selection of target genes becomes both critical and complex. Effective design requires not only consolidating what is already known about the biology underlying a phenotype of interest, but also hypothesizing about genes that could generate new, previously unrecognized insights. This latter task relies on abductive reasoning—the process of proposing plausible explanations that extend beyond existing evidence—which poses a particular challenge for large language models (LLMs). We compared three modes of LLM-assisted target selection. Out-of-the-box LLMs are effective at reproducing well-established biology. LLMs augmented with literature retrieval extend this capacity to integrate more obscure or rarely cited findings. Agentic systems equipped with biological foundation models go a step further, capturing latent biological structure and supporting abductive reasoning by proposing candidate genes not yet described in the literature. Benchmarking these approaches against whole-genome perturbation screens highlights distinct strengths and weaknesses in compiling comprehensive gene sets, with trade-offs in explainability, novelty, and cost. Together, these results provide a comparative framework for integrating AI systems into the design of large-scale perturbation experiments.

Contrastive multimodal embeddings link chromatin accessibility and CNV to expression in metastatic prostate cancer

Siddharth Mahesh

Poster 76

Single-cell multi-omic datasets promise cell-state-resolved mechanisms yet remain difficult to model due to extreme dimensionality, sparsity, and noise across assays. We present a modality-aware deep learning framework that learns interpretable joint representations while respecting each modality's unique structure. The workflow entails a BERT transformer for RNA, and nested autoencoders for chromatin accessibility and copy-number variation (CNV), aligning them via contrastive learning into a shared latent space. We demonstrate the framework on paired single-nucleus ATAC-seq and RNA-seq, with CNV inferred from ATAC-seq, in a cohort of 52 metastatic castration-resistant prostate cancer (mCRPC) samples from 23 individuals. For interpretability, we apply integrated gradients to each encoder to obtain pathway-level attributions and transcription factor-gene relationships, benchmarking these against a priori bulk-derived mCRPC-subtype-specific signatures as controls. The modality-specific encoders recover canonical genomic, chromatin, and CNV signatures across mCRPC subtypes established in the bulk literature, while the joint latent space links the three modalities to identify cell-state-specific regulons across mCRPC subtypes. Methodologically, the contributions are threefold: (i) modality-tailored architecture choices that learn patient-invariant representations; (ii) attribution-driven in silico validation against a priori knowledge that evaluates biological plausibility and reduces “black-box” concerns; and (iii) a multimodal embedding suitable for downstream tasks such as subtype mapping, perturbation reasoning, and cross-modal mechanistic inference. Although utilized in mCRPC, the framework is disease-agnostic. While still a work-in-progress, the learned representations can couple with generative models to simulate transcriptional consequences of arbitrary CNV-chromatin configurations at subclonal resolution, enabling in silico hypothesis generation prior to experimental validation.

Deciphering Regulatory Mechanisms in Pediatric Leukemia with Single-Cell Multiomics and Deep Learning

Seungbyn Baek
Poster 77

Pediatric leukemia remains a leading cause of childhood cancer mortality, yet progress is constrained by limited insight into how genetic lesions derail hematopoiesis. Master regulators such as lineage-specific transcription factors (TFs) and TF fusions drive leukemic arrest. The low mutational complexity of pediatric leukemias makes them tractable for mechanistic dissection. However, beyond driver TF fusions, the regulatory mechanisms including secondary hits and noncoding variants that shape phenotype remain poorly defined. This gap limits precise patient stratification and the development of targeted therapies. In this study, we generated single-cell multiome (RNA+ATAC) profiles from 22 index cases spanning diverse pediatric leukemia subtypes. Using interpretable sequence-to-profile deep learning models (ChromBPNet), we constructed base-resolution maps of cis-regulatory syntax that capture TF motif grammar (presence, spacing, orientation, cooperativity) and provide predictive scores quantifying the impact of base- or motif-level changes on chromatin accessibility and gene expression. Preliminary models across 22 cases recover subtype-specific TF motifs at base resolution. Next, we will leverage these models to connect syntax to gene programs in joint ATAC+RNA profiles and interpret the function of recurrent noncoding variants using in-silico mutagenesis.

Re/Composer Allows for the Discovery of Higher Order Mechanisms of Context Dependency That Facilitate Intestinal Regeneration in a New Spatial Transcriptomics Atlas of Regeneration

Shyam Patel

Poster 78

How tissue and cell state dynamics orchestrate intestinal tissue regeneration remains ill-defined. Applying the 10X Genomics Xenium platform allowed us to generate a high resolution spatial transcriptomic map of the regenerating murine small intestine at multiple timepoints between 0 and 9 days post injury (dpi). In the intestine, the stereotypic tissue organizing unit is the crypt-villus unit (CVU). To assess tissue-scale dynamics I developed an algorithm I call Re/Composer that integrates single cell profiles from our spatial transcriptomic atlas into CVUs. Applying Re/Composer recapitulated canonical dynamics—such as shifts in Lgr5 intestinal stem cells (ISCs) and the emergence of Clu revival stem cells (RevSCs) – and novel discrete changes in the stroma associated with each CVU class. Stratifying celltype dynamics by CVU class revealed the avenues by which the intestine imposes context dependency to affect cell state and CVU organization to drive successful regeneration. Our framework demonstrates how leveraging repeating tissue units can integrate biological architecture with spatial transcriptomics, providing a scalable strategy for dissecting regeneration and other dynamic tissue processes.

Single-cell long-read transcriptomics reveals isoform centric molecular features of rare pseudomyxoma peritonei disease

Kyungtae Lee
Poster 79

Pseudomyxoma peritonei (PMP) is a rare neoplastic condition characterized by accumulation of mucin and widespread dissemination of tumor cells, commonly originated from appendiceal mucinous neoplasms (AMN). Our research group previously published the first single-cell transcriptome analysis of 14 AMN and PMP patient, revealing molecular signatures of tumor epithelia cells and tumor infiltrating lymphocytes. To further investigate AMN and PMP tumors at isoform level, we conducted single-cell long-read sequencing of 10 biopsies from adjacent normal tissue, AMN and PMP tumor from 5 patients using Oxford Nanopore Technologies. We identified 35,095 expressed genes and 130,272 transcripts, including 50,330 that were novel transcripts. Long-read data and derived isoform expression were demonstrated to effectively capture full-length RNA molecules and cellular heterogeneity. Comparison between adjacent normal and AMN samples revealed stronger goblet cell differentiation and proliferation signatures in AMN. Notably, 29 isoforms in 21 genes such as MUC2, REG4, SPINK4 and TIMP1 were specifically expressed in AMN, implying their potential as early diagnostic isoform markers. We also identified 20 significantly and differentially expressed isoforms with gene expression changes, indicating alternative splicing regulation independent of overall gene expression. Our ongoing research aims to identify neoantigens derived from patient-specific mutations and alternative splicing events, with the goal of uncovering novel drivers of AMN and PMP tumor biology at single-cell isoform resolution.

TorchDR: Scalable GPU-Accelerated Dimensionality Reduction

Hugues Van Assel

Poster 80

Exploring high-dimensional data requires dimensionality reduction (DR) methods that are both fast and scalable. TorchDR is a GPU-accelerated PyTorch library that delivers orders of magnitude speedups over traditional CPU-based implementations, enabling practitioners to iterate rapidly on their data, gain insights quickly, and seamlessly integrate into modern ML workflows. Designed for researchers and practitioners alike, TorchDR combines high performance with ease of use: a consistent API, native GPU acceleration, and compatibility with torch.compile make it a practical solution for experimenting with multiple DR methods at scale.

TorchDR integrates a comprehensive family of neighbor embedding algorithms, such as t-SNE, UMAP, LargeVis, PaCMAP, and related variants, alongside spectral methods like PCA, KernelPCA, and PHATE. Built on the PyTorch ecosystem, the library leverages advanced backends such as FAISS for approximate nearest neighbors and KeOps for symbolic, memory efficient operations, providing a unified, modular, and extensible framework for large scale DR.

Crucially, TorchDR introduces multi GPU support, where k nearest neighbor graphs are cleverly sharded across devices to achieve giga scale embeddings in minutes. This capability redefines what is possible in DR, enabling the analysis of datasets with tens of millions of points, now critical in frontier applications such as molecular discovery and single cell biology. By combining scalability, flexibility, and cutting edge GPU acceleration, TorchDR establishes a new standard for dimensionality reduction at scale.

From Photosynthetic Dimers to Artificial Light-Harvesting: A Theoretical Framework for Coherence Protection

Xinyan Song
Poster 81

Efficient energy transfer is a hallmark of photosynthesis, yet its underlying quantum mechanisms remain debated. We propose a theoretical simulation using an exciton dimer model to investigate vibronic–exciton coherence protection. Combining Lindblad dynamics with hierarchical equations of motion (HEOM), we simulate the asymmetric dissipation of symmetric and antisymmetric vibrational modes and design numerical experiments of simplified two-dimensional electronic spectroscopy and dynamic Stokes shift to verify the role of phase synchronization in extending coherence lifetimes. The framework can be extended to multi-pigment networks and cavity-QED settings, offering theoretical foundations for artificial light-harvesting and quantum device design.

scDORI: Scalable Deep Learning of Enhancer-Gene Networks from Single-Cell Multi-Omics

Manu Saraswat

Poster 82

Inferring enhancer-mediated gene regulatory networks (eGRNs) from single-cell multi-omics remains challenging at scale, particularly for modeling activation and repression across continuous cell states. Current methods are limited by scalability, reliance on discrete annotations, and weak repression modeling. We developed scDoRI (single-cell Deep multi-Omic Regulatory Inference), a deep learning framework that infers eGRNs from joint RNA-seq and ATAC-seq data. scDoRI uses label-free topic modeling to decompose multi-omic profiles into regulatory “topics” linking transcription factors, cis-regulatory elements, and target genes under mechanistic constraints. Multiple coupled decoders capture peak co-accessibility, TF binding, and expression correlations. Cells are represented as mixtures of topics, enabling continuous modeling of activation and repression while scaling to millions of cells. Applied to >1M nuclei from twelve glioblastoma patients, scDoRI revealed a structured hierarchy of malignant states with distinct regulatory topics. Neuronal-like cells emerged as uniquely stable, actively repressing alternate fates, whereas progenitor- and hypoxia-like states remained permissive and highly plastic. Quantifying topic activation potential and repression scores provided directional maps of state transitions, highlighting dynamic flows between progenitor-like, astrocytic-like, and hypoxic states while identifying neuronal-like cells as terminally restricted. From these programs, scDoRI identified MYT1L as a master repressor safeguarding neuronal-like identity. Perturbation experiments validated this prediction: MYT1L overexpression silenced alternate chromatin programs, reduced proliferation and invasion, and extended survival in vivo, while loss of function reactivated plasticity and accelerated malignancy. scDoRI is openly available on GitHub and generalizes to diverse multiome datasets, offering a scalable strategy to dissect cell identity, predict state transitions, and uncover therapeutic vulnerabilities in heterogeneous systems.

Community-based biomedical context to unlock agentic systems

Darius Schaub

Poster 83

Large language models (LLMs) face reliability challenges stemming from hallucinations and insufficient access to validated scientific resources. Existing solutions are often fragmented and limited to specific applications, hindering broader adoption and interoperability. Here, we present Biomedical Context for Artificial Intelligence (BioContextAI), an open-source initiative centered on Model Context Protocol (MCP) servers to address these limitations. BioContextAI provides a community-oriented registry for discovering domain-specific MCP servers and a proof-of-concept server implementation that integrates widely-used biomedical knowledgebases. By enabling standardized access to validated scientific knowledge, BioContextAI aims to facilitate the development of composable agentic systems for biomedical research. Together, this work contributes to an emerging ecosystem of community-driven approaches for expanding the capabilities and reliability of biomedical AI systems.

SIMBA+: Dissecting genetic variant function through single-cell multiomics integration

Junxi Feng
Poster 84

Translating genome-wide association studies (GWAS) results into variant mechanism remains challenging. We present SIMBA+, a probabilistic graph framework that integrates single-cell multiomics with GWAS to identify variant target genes and cellular contexts. SIMBA+ constructs a unified knowledge graph linking cells, genes, and regulatory elements, and learns node embeddings to score peak-cell-gene relationships via metapath analysis. We showcase SIMBA+ on GWAS variants associated with atopic dermatitis, revealing how disease variants regulate immune genes like CSF2RB specifically in monocytes and NLRP3 in CD14+ populations. Systematic benchmarking demonstrates superior performance over existing methods, with higher enrichment for eQTLs and CRISPR-validated regulatory links, particularly for distal and cell-type specific interactions. Beyond link prediction, SIMBA+ uniquely enables heritability decomposition at single-cell resolution through factor analysis. Applying SIMBA+ to 75 complex traits across three multiome atlases spanning blood, bone marrow, and multiple tissues, we identified disease-relevant cell states among hundreds of cell types, discovered regulatory programs missed by pseudobulk approaches, and provided the first single-cell resolution heritability estimates. By providing variant interpretation at single-cell resolution, SIMBA+ advances translation of GWAS findings into biological insights.

Celldega: Toolkit for Analysis and Visualization of Spatial and Single-Cell Data

Huan Wang
Poster 85

Spatial transcriptomics (ST) combines high-dimensional single-cell gene expression with traditional microscopy imaging, transforming our understanding of cellular states, communication, and tissue organization. However, open-source ST visualization tools are unable to scale to large datasets (beyond 100 million transcripts), while closed-source commercial tools come with high costs and limited extensibility. We present Celldega, an open-source Python and JavaScript library that integrates seamlessly with Scverse workflows. Celldega provides scalable, feature-rich visual exploration of both spatial and high-dimensional data, supports novel neighborhood analyses of ST data (e.g., alpha shape-based tissue habitats), and enables interactive visualization in notebooks and web galleries. Using modern web technologies (e.g., deck.gl) and vector tiling with Parquet and Apache Arrow, Celldega efficiently stores and streams large datasets from local, cloud or public GitHub/HuggingFace repositories. We demonstrate Celldega within Scanpy/Squidpy workflows for exploring both spatial and high-dimensional data, and showcase how it can identify tissue neighborhoods (e.g., cell type population niches and overlapping cell type alpha-shape habitats). Finally, we utilize Celldega to visually explore 3D reconstructions of the developing mouse cranium comprising >4 million single cells with full-transcriptome expression from Illumina Spatial Technology data. Celldega extends Scverse by integrating scalable visualization with novel spatial analysis.

Glass-Box UMAP

James Golden

Poster 86

UMAP is a powerful tool for visualizing high-dimensional datasets, but the black-box nature of its highly nonlinear mapping function obscures how it uses input features to form distinctive clusters. We enable exact interpretation of the mapping function through an alternative implementation of parametric UMAP by learning the embedding with a deep network that is locally linear (but globally nonlinear) with respect to the input features. This “glass-box” approach provides a set of exact feature contributions as linear weights that determine the embedding of each data point. By computing the exact feature contribution for each point in a dataset, we can directly quantify which features are responsible for forming each cluster. We explore the feature contributions from this glass-box augmentation of UMAP with several datasets and compare them with features found at the cluster level by differential expression.

Study impact of human marijuana use on peripheral immune cells from HIV infected, and ART suppressed people by CITEseq multi-omics analysis

Urja Bhatt

Poster 87

Despite effective control of human immunodeficiency virus-1 (HIV-1) replication on anti-retroviral therapies (ART), HIV associated chronic inflammation is a major cause of co-morbidities in people with HIV (PWH). Immune dysfunction from HIV disease has been recognized as a cause of this chronic inflammation. To relieve pain, recreational and medicinal cannabis (CB) use has increased significantly among PWH. Previous research has shown cannabinoids to be anti-inflammatory in PWH, however, specific molecular mechanisms that uniquely address its effects on the complex immune milieu in PWH have not been formally assessed. To identify the short-term effects of CB on immune cells from PWH, we performed single-cell gene expression on PBMCs (n=3) exposed to CB ex-vivo for 6 hours compared to a non-HIV infected control. Further, to identify molecular signatures associated with long-term CB use in different immune cell types, we performed Cellular Indexing of Transcriptomes and Epitopes by Sequencing (CITE-seq) on PBMCs from a cohort of ART-suppressed PWH (n=27) with confirmed plasma toxicology for CB compared to non-CB using PWH on ART as control. Data analysis was performed using scanpy, muon and MOFA+ packages. We found altered frequencies of different immune cell subsets among CB using and non-CB using PWH. Also, inflammatory pathways were downregulated in CB using samples. This study uncovers pathways responsible for immunomodulatory effects of CB in PWH at an unprecedented resolution.

